

# CREDIT CARD APPROVAL RECORD

PRISCILLA YOU



# ABOUT DATASET

## **credit card approval**



The dataset contains multiple variables, including demographic details, financial attributes, which influence the approval decision.

The project focuses on analyzing correlations among these variables, such as income level, age, education type, occupation type, to uncover patterns that contribute to credit card approval or rejection.

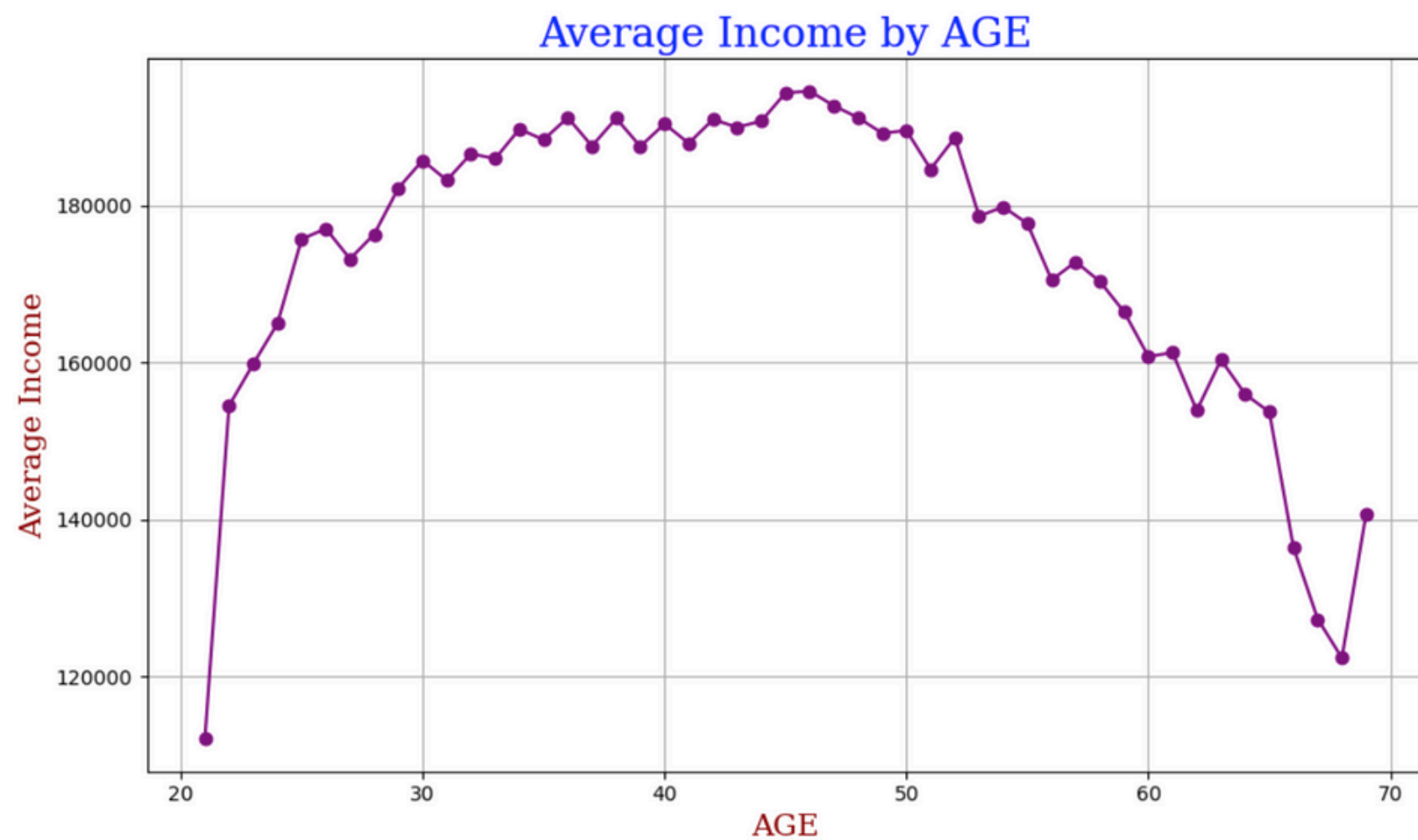
| <div><div></div><div></div></div> <div>123 ID</div> <div><div></div><div></div></div> | <div><div></div><div></div></div> <div>CODE_GENDER</div> <div><div></div><div></div></div> | <div><div></div><div></div></div> <div>FLAG_OWN_CAR</div> <div><div></div><div></div></div> | <div><div></div><div></div></div> <div>FLAG_OWN_REALTY</div> <div><div></div><div></div></div> | <div><div></div><div></div></div> <div>123 CNT_CHILDREN</div> <div><div></div><div></div></div> | <div><div></div><div></div></div> <div>123 AMT_INCOME_TOTAL</div> <div><div></div><div></div></div> |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 05008804                                                                              | M                                                                                          | Y                                                                                           | Y                                                                                              | 0                                                                                               | 427500.0                                                                                            |
| 15008805                                                                              | M                                                                                          | Y                                                                                           | Y                                                                                              | 0                                                                                               | 427500.0                                                                                            |
| 25008806                                                                              | M                                                                                          | Y                                                                                           | Y                                                                                              | 0                                                                                               | 112500.0                                                                                            |
| 35008808                                                                              | F                                                                                          | N                                                                                           | Y                                                                                              | 0                                                                                               | 270000.0                                                                                            |
| 45008809                                                                              | F                                                                                          | N                                                                                           | Y                                                                                              | 0                                                                                               | 270000.0                                                                                            |
| 55008810                                                                              | F                                                                                          | N                                                                                           | Y                                                                                              | 0                                                                                               | 270000.0                                                                                            |
| 65008811                                                                              | F                                                                                          | N                                                                                           | Y                                                                                              | 0                                                                                               | 270000.0                                                                                            |
| 75008812                                                                              | F                                                                                          | N                                                                                           | Y                                                                                              | 0                                                                                               | 283500.0                                                                                            |
| 85008813                                                                              | F                                                                                          | N                                                                                           | Y                                                                                              | 0                                                                                               | 283500.0                                                                                            |
| 95008814                                                                              | F                                                                                          | N                                                                                           | Y                                                                                              | 0                                                                                               | 283500.0                                                                                            |
| 105008815                                                                             | M                                                                                          | Y                                                                                           | Y                                                                                              | 0                                                                                               | 270000.0                                                                                            |

| <div><div></div><div></div></div> <div>NAME_INCOME_TYPE</div> <div><div></div><div></div></div> | <div><div></div><div></div></div> <div>NAME_EDUCATION_TYPE</div> <div><div></div><div></div></div> | <div><div></div><div></div></div> <div>NAME_FAMILY_STATUS</div> <div><div></div><div></div></div> | <div><div></div><div></div></div> <div>NAME_HOUSING_TYPE</div> <div><div></div><div></div></div> | <div><div></div><div></div></div> <div>123 DAYS_BIRTH</div> <div><div></div><div></div></div> |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Working                                                                                         | Higher education                                                                                   | Civil marriage                                                                                    | Rented apartment                                                                                 | -12005                                                                                        |
| Working                                                                                         | Higher education                                                                                   | Civil marriage                                                                                    | Rented apartment                                                                                 | -12005                                                                                        |
| Working                                                                                         | Secondary / secondary special                                                                      | Married                                                                                           | House / apartment                                                                                | -21474                                                                                        |
| Commercial associate                                                                            | Secondary / secondary special                                                                      | Single / not married                                                                              | House / apartment                                                                                | -19110                                                                                        |
| Commercial associate                                                                            | Secondary / secondary special                                                                      | Single / not married                                                                              | House / apartment                                                                                | -19110                                                                                        |
| Commercial associate                                                                            | Secondary / secondary special                                                                      | Single / not married                                                                              | House / apartment                                                                                | -19110                                                                                        |
| Commercial associate                                                                            | Secondary / secondary special                                                                      | Single / not married                                                                              | House / apartment                                                                                | -19110                                                                                        |
| Pensioner                                                                                       | Higher education                                                                                   | Separated                                                                                         | House / apartment                                                                                | -22464                                                                                        |
| Pensioner                                                                                       | Higher education                                                                                   | Separated                                                                                         | House / apartment                                                                                | -22464                                                                                        |
| Pensioner                                                                                       | Higher education                                                                                   | Separated                                                                                         | House / apartment                                                                                | -22464                                                                                        |
| Working                                                                                         | Higher education                                                                                   | Married                                                                                           | House / apartment                                                                                | -16872                                                                                        |

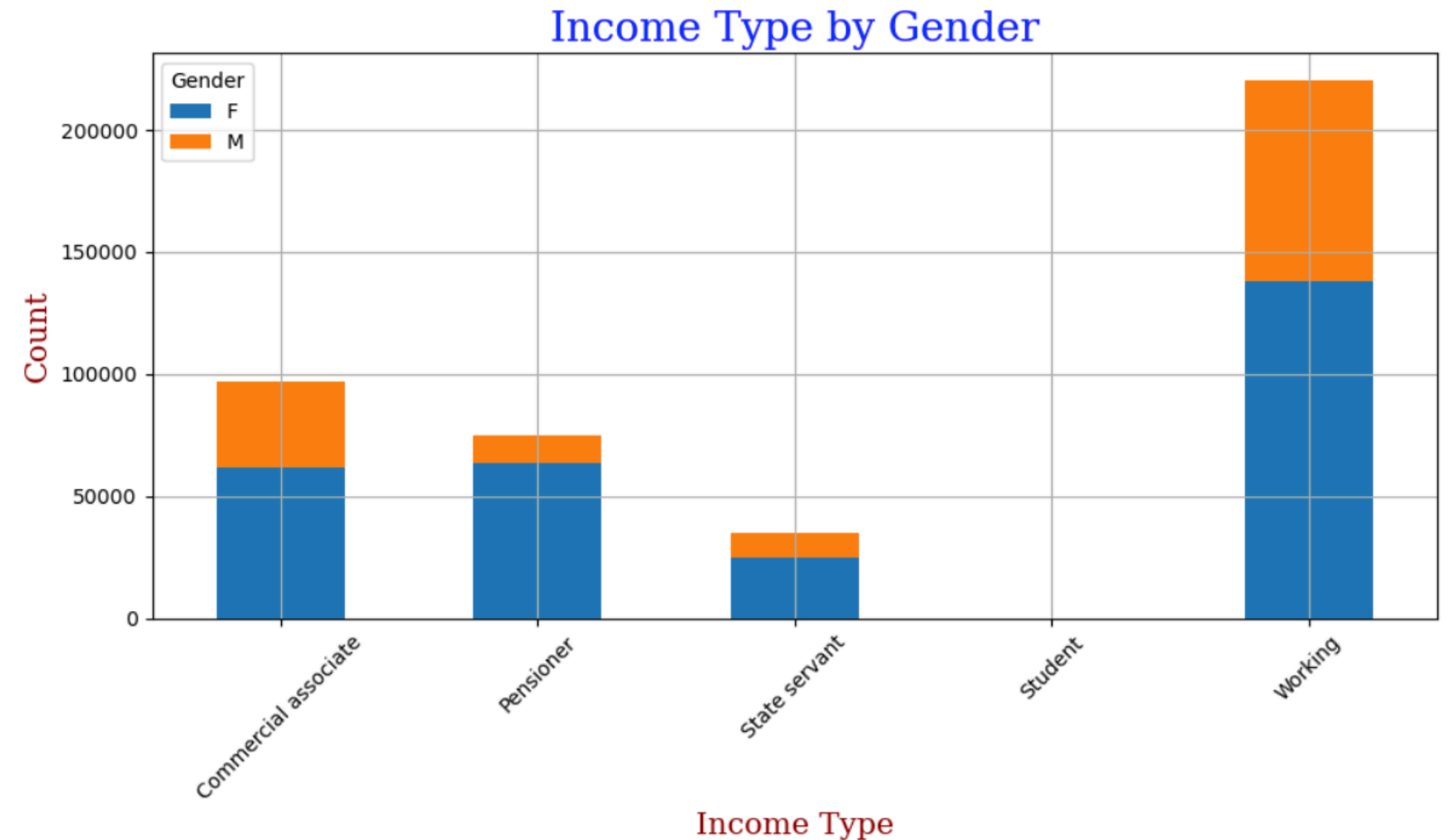
| <div><div></div><div></div></div> <div>123 FLAG_MOBIL</div> <div><div></div><div></div></div> | <div><div></div><div></div></div> <div>123 FLAG_WORK_PHONE</div> <div><div></div><div></div></div> | <div><div></div><div></div></div> <div>123 FLAG_PHONE</div> <div><div></div><div></div></div> | <div><div></div><div></div></div> <div>123 FLAG_EMAIL</div> <div><div></div><div></div></div> | <div><div></div><div></div></div> <div>OCCUPATION_TYPE</div> <div><div></div><div></div></div> | <div><div></div><div></div></div> <div>123 CNT_FAM_MEMBERS</div> <div><div></div><div></div></div> |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 1                                                                                             | 1                                                                                                  | 0                                                                                             | 0                                                                                             | Unknown                                                                                        | 2.0                                                                                                |
| 1                                                                                             | 1                                                                                                  | 0                                                                                             | 0                                                                                             | Unknown                                                                                        | 2.0                                                                                                |
| 1                                                                                             | 0                                                                                                  | 0                                                                                             | 0                                                                                             | Security staff                                                                                 | 2.0                                                                                                |
| 1                                                                                             | 0                                                                                                  | 1                                                                                             | 1                                                                                             | Sales staff                                                                                    | 1.0                                                                                                |
| 1                                                                                             | 0                                                                                                  | 1                                                                                             | 1                                                                                             | Sales staff                                                                                    | 1.0                                                                                                |
| 1                                                                                             | 0                                                                                                  | 1                                                                                             | 1                                                                                             | Sales staff                                                                                    | 1.0                                                                                                |
| 1                                                                                             | 0                                                                                                  | 1                                                                                             | 1                                                                                             | Sales staff                                                                                    | 1.0                                                                                                |
| 1                                                                                             | 0                                                                                                  | 0                                                                                             | 0                                                                                             | Unknown                                                                                        | 1.0                                                                                                |
| 1                                                                                             | 0                                                                                                  | 0                                                                                             | 0                                                                                             | Unknown                                                                                        | 1.0                                                                                                |
| 1                                                                                             | 0                                                                                                  | 0                                                                                             | 0                                                                                             | Unknown                                                                                        | 1.0                                                                                                |

| Feature name        | Type        | Explanation              |
|---------------------|-------------|--------------------------|
| ID                  | Categorical | Client number            |
| CODE_GENDER         | Categorical | Gender                   |
| FLAG_OWN_CAR        | Categorical | Is there a car           |
| FLAG_OWN_REALTY     | Categorical | Is there a property      |
| CNT_CHILDREN        | Numerical   | Number of children       |
| AMT_INCOME_TOTAL    | Numerical   | Annual income            |
| NAME_INCOME_TYPE    | Categorical | Income category          |
| NAME_EDUCATION_TYPE | Categorical | Education level          |
| NAME_FAMILY_STATUS  | Categorical | Marital status           |
| NAME_HOUSING_TYPE   | Categorical | Way of living            |
| DAYS_BIRTH          | Numerical   | Birthday                 |
| DAYS_EMPLOYED       | Numerical   | Start date of employment |
| FLAG_MOBIL          | Categorical | Is there a mobile phone  |
| FLAG_WORK_PHONE     | Categorical | Is there a work phone    |
| FLAG_PHONE          | Categorical | Is there a phone         |
| FLAG_EMAIL          | Categorical | Is there an email        |
| OCCUPATION_TYPE     | Categorical | Occupation               |
| CNT_FAM_MEMBERS     | Numerical   | Family size              |

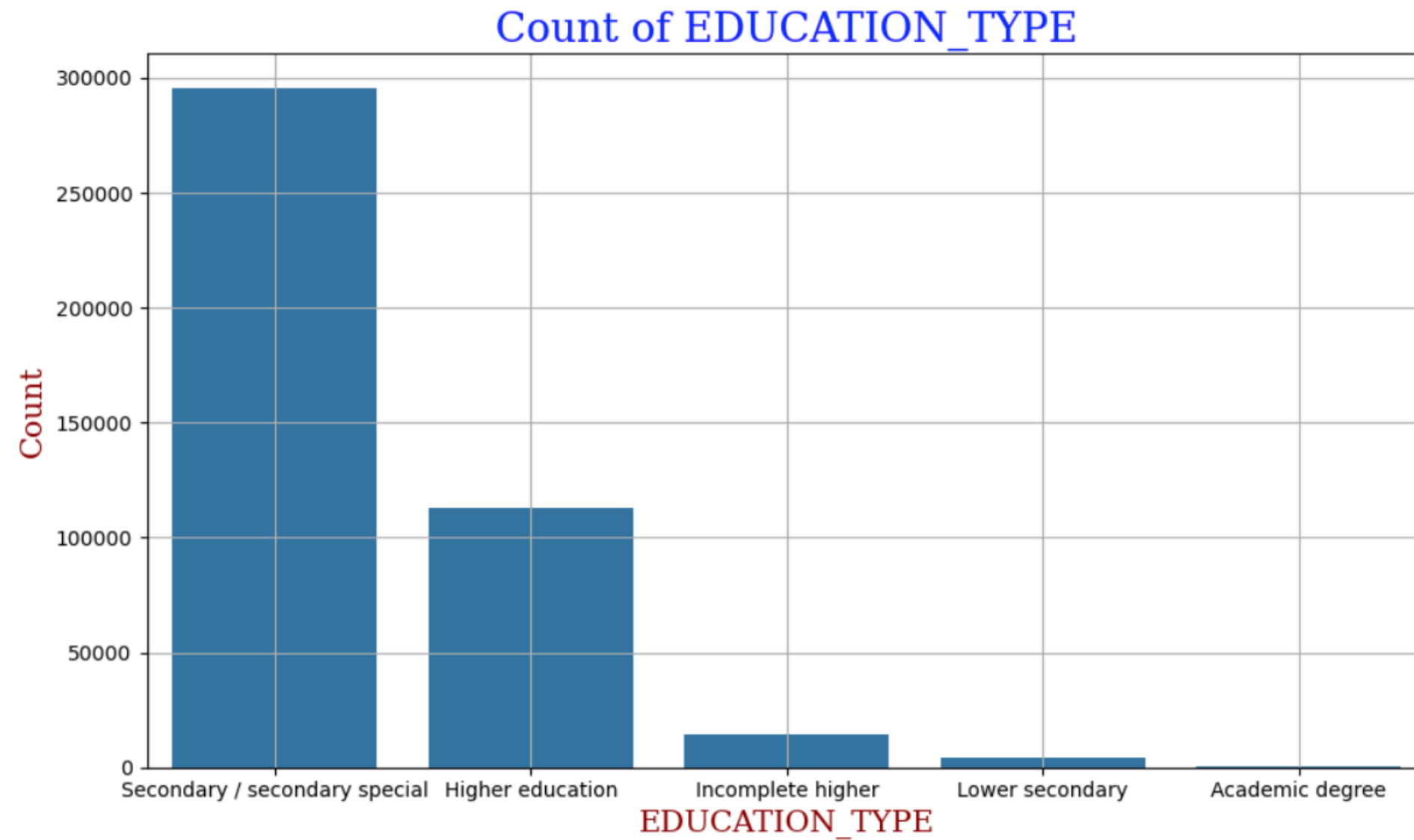
# PHASE 1



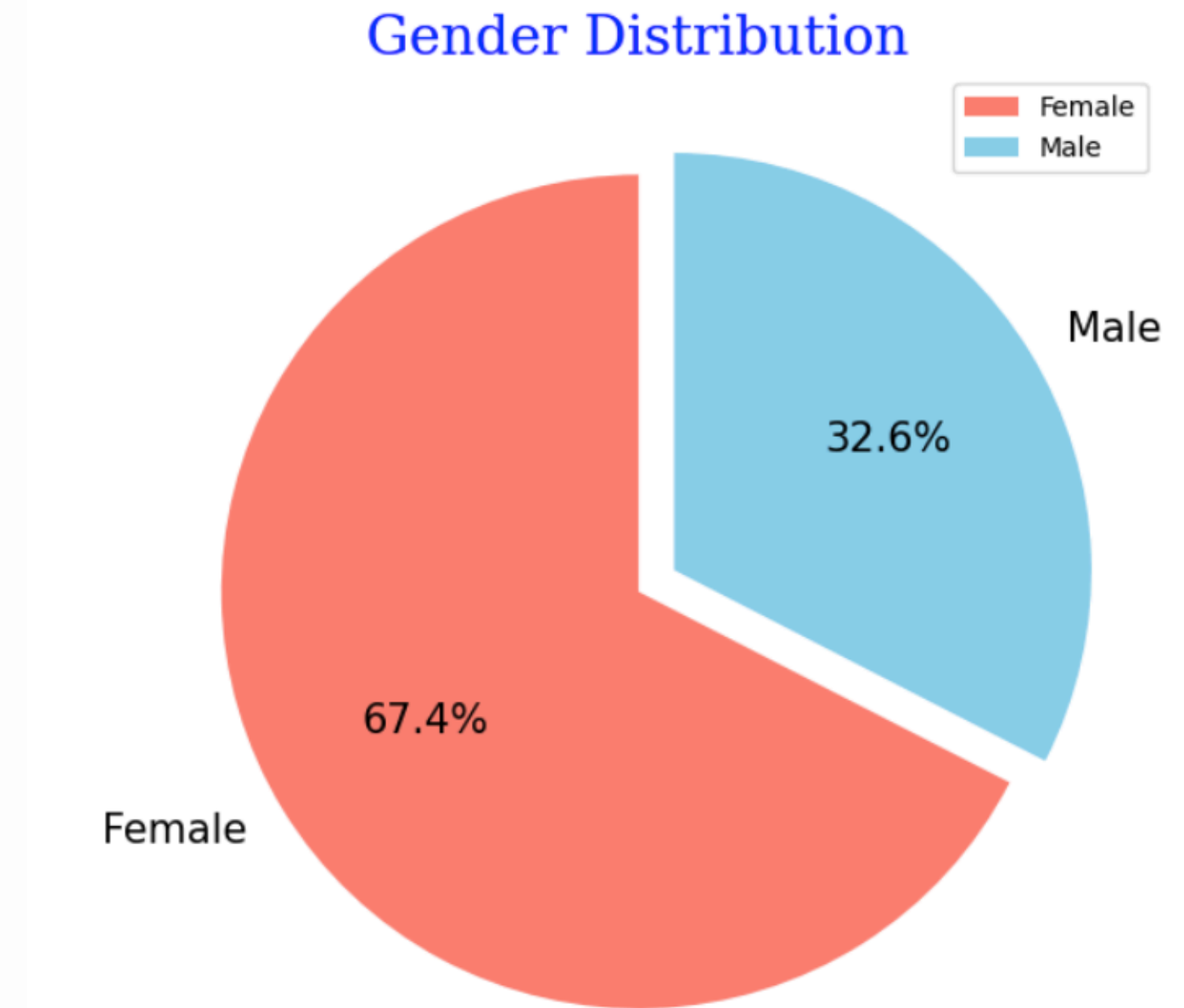
Individuals aged 40 to 50 have the highest average income



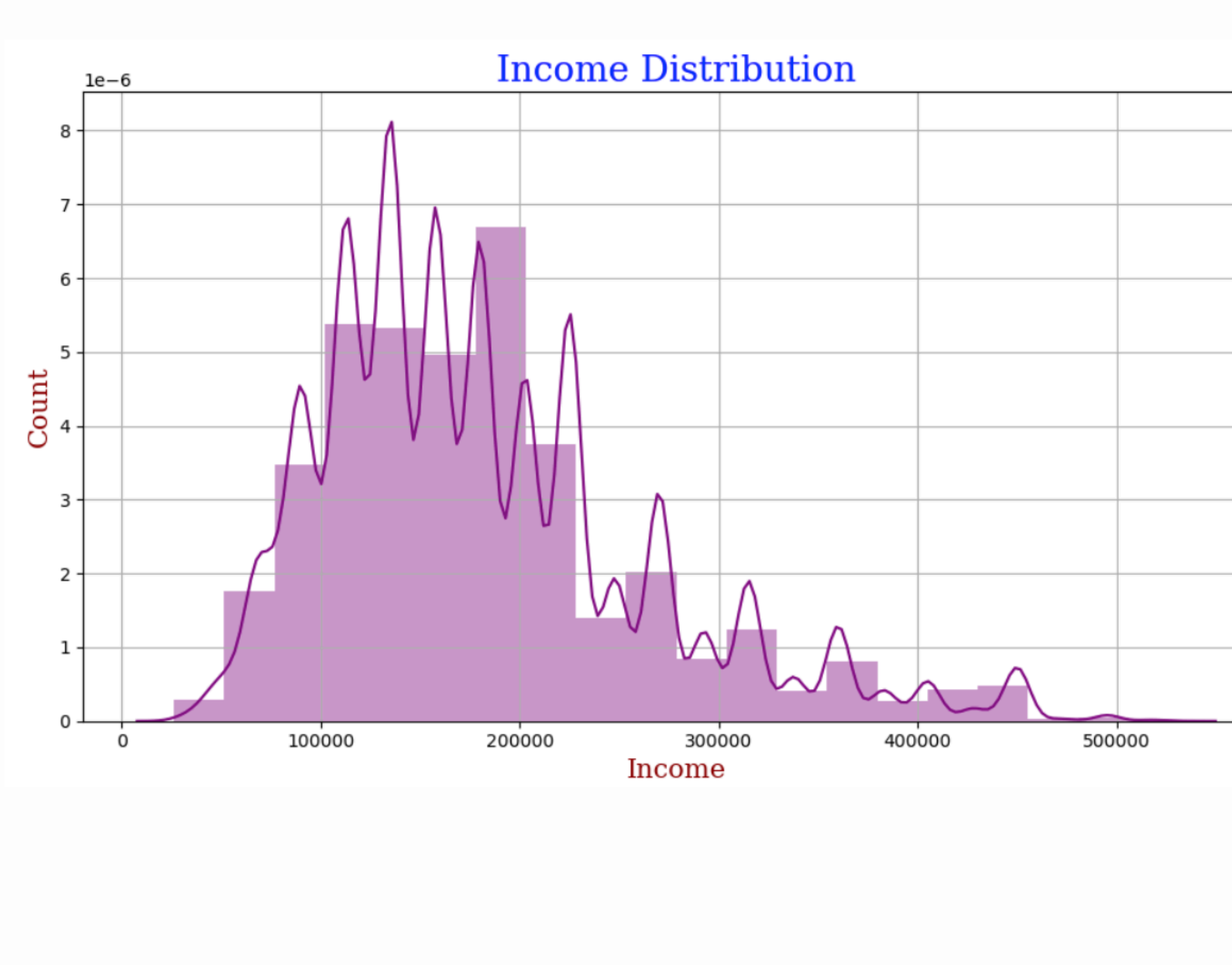
The "Working" category is the most prominent for both males and females



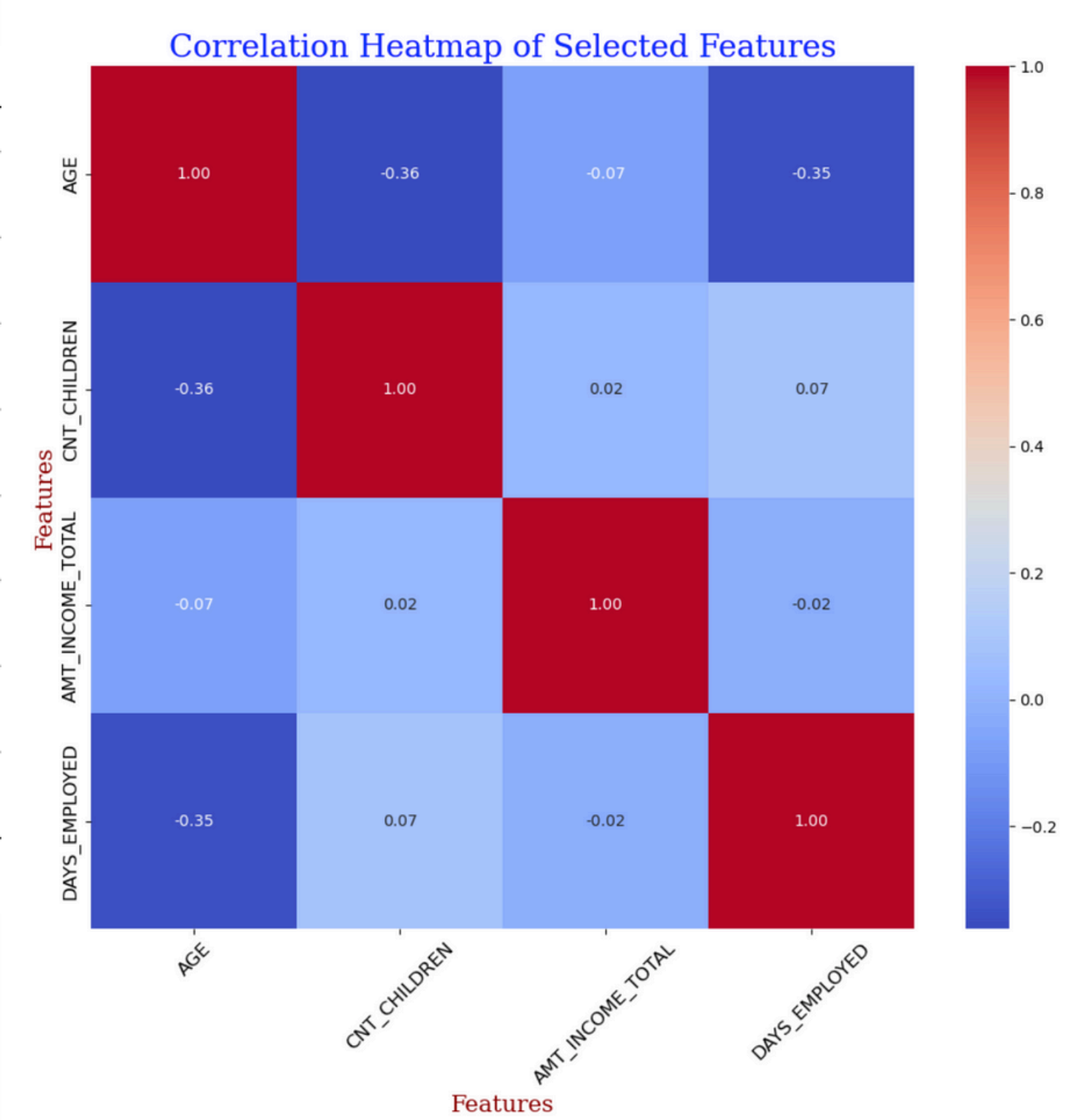
The majority of individuals have a "Secondary/Secondary Special" education



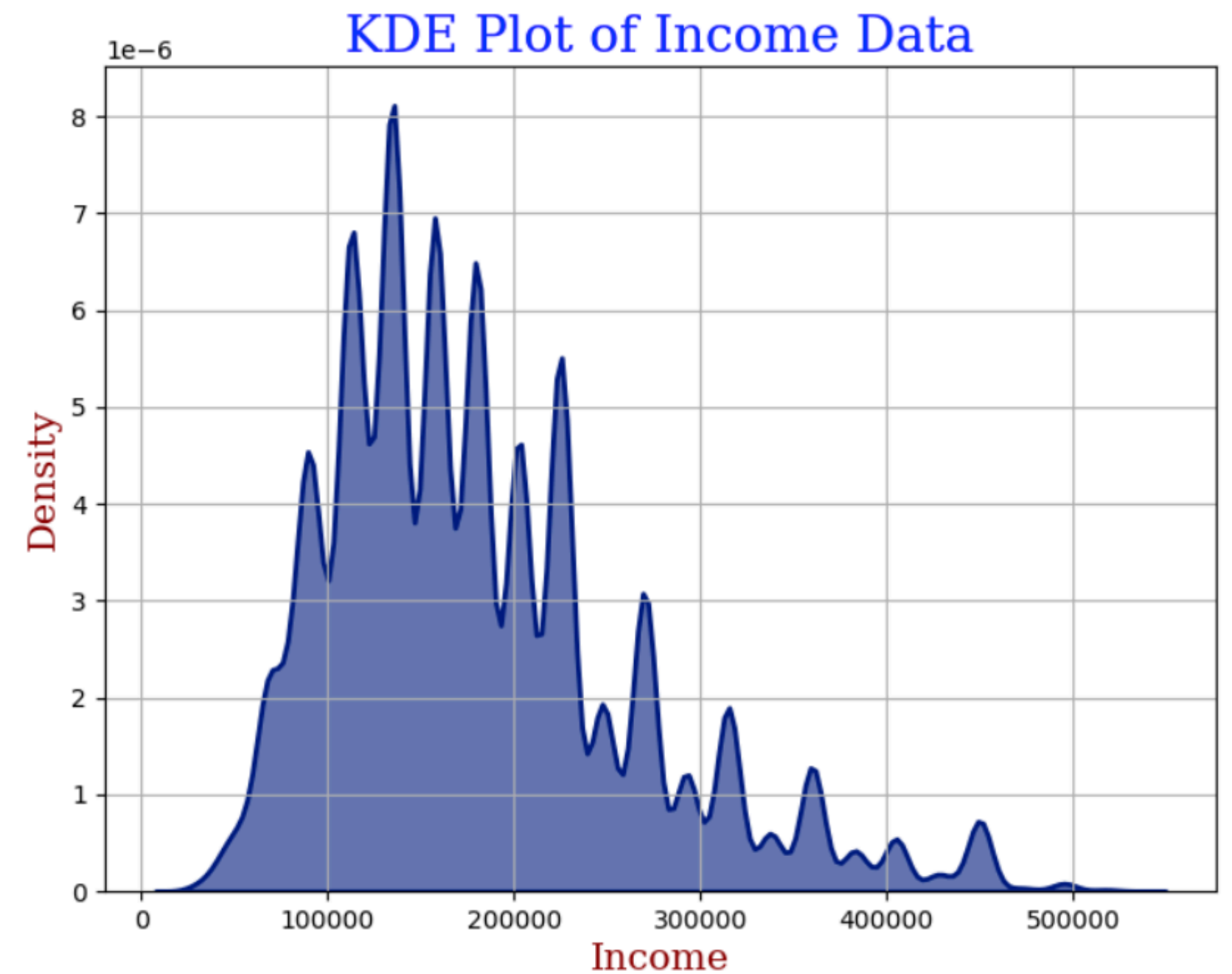
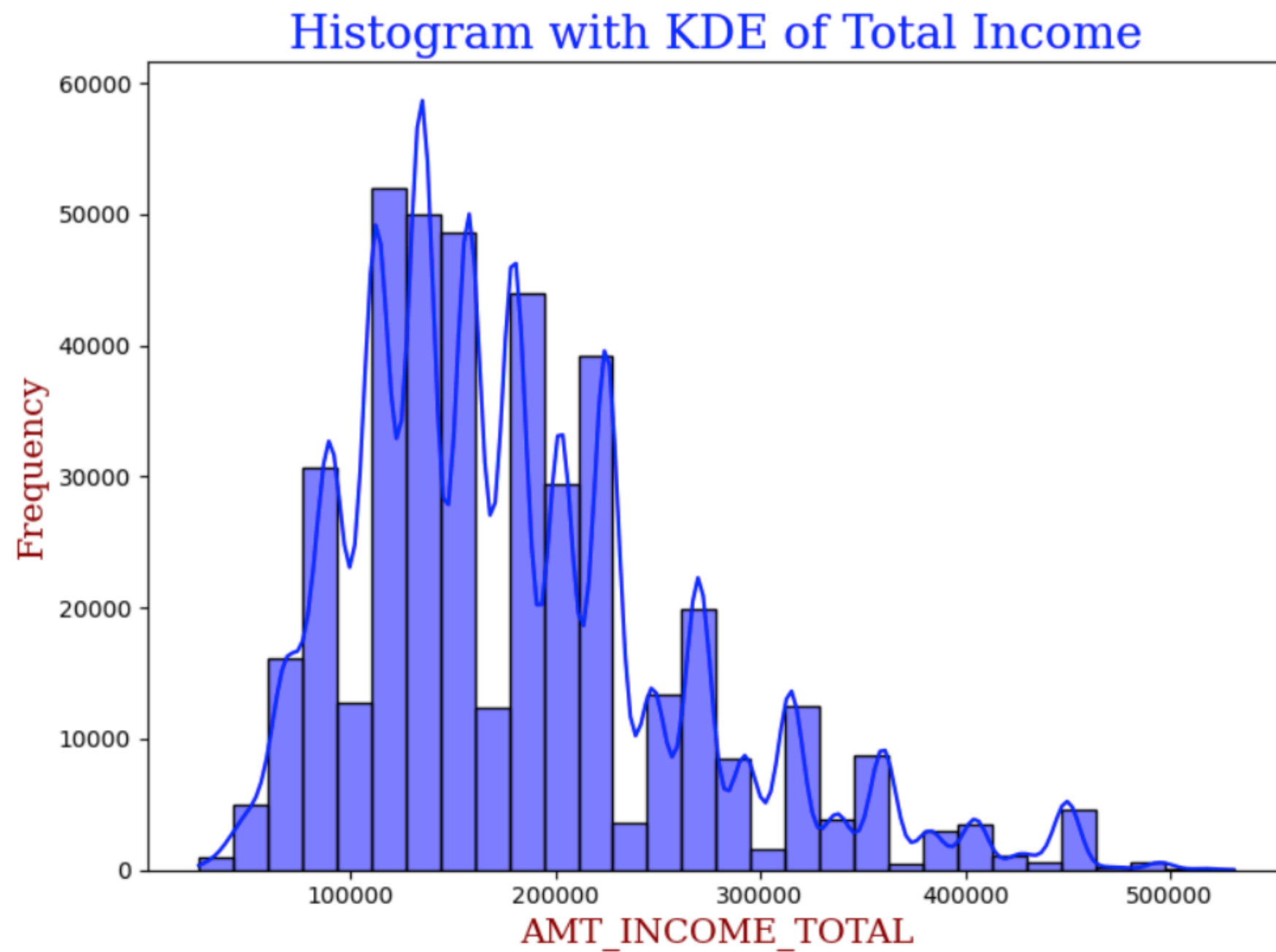
Females constitute 67.4% and Males 32.6% of the dataset



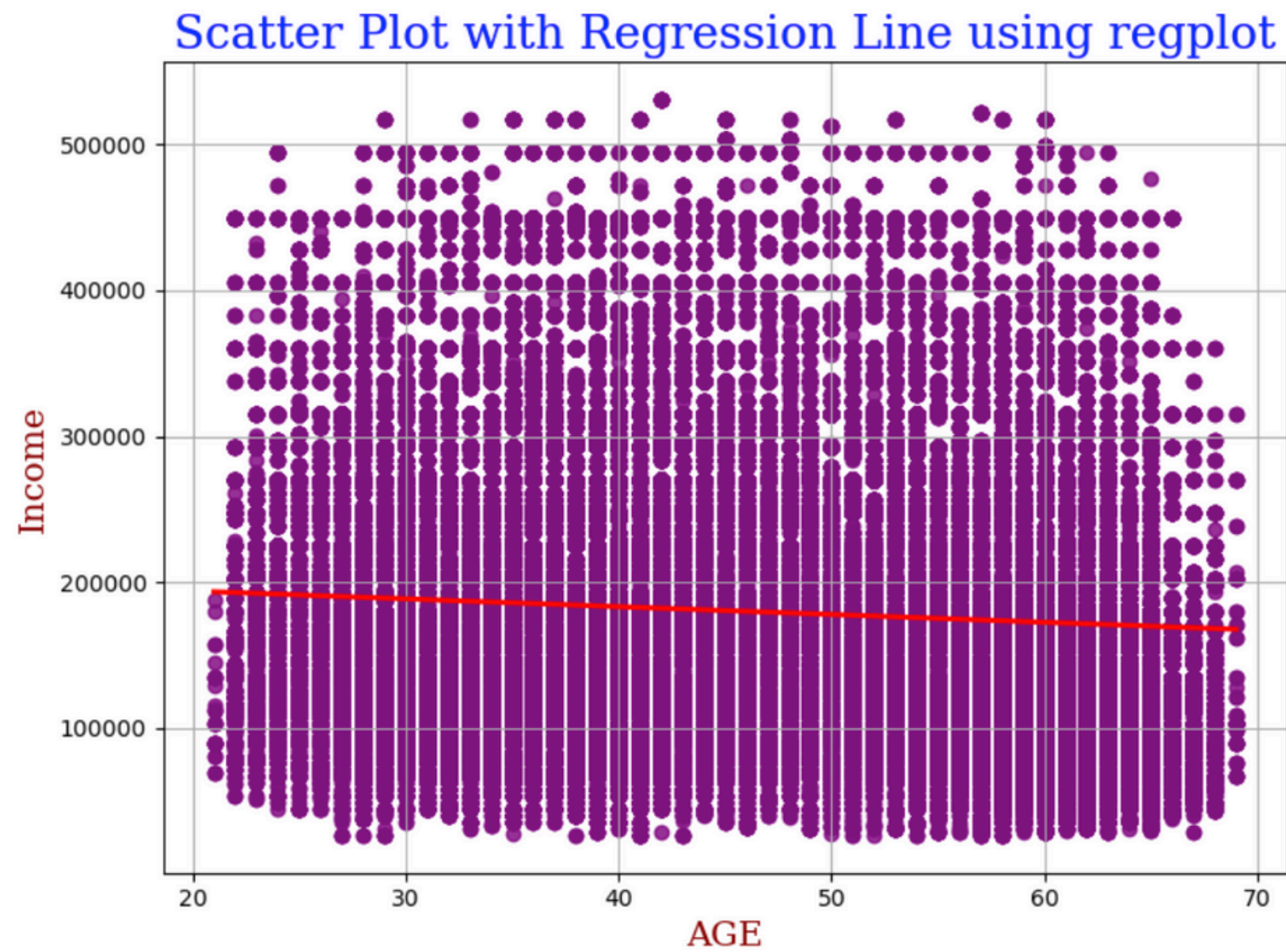
The data is right-skewed, with most individuals earning between 100,000 and 300,000



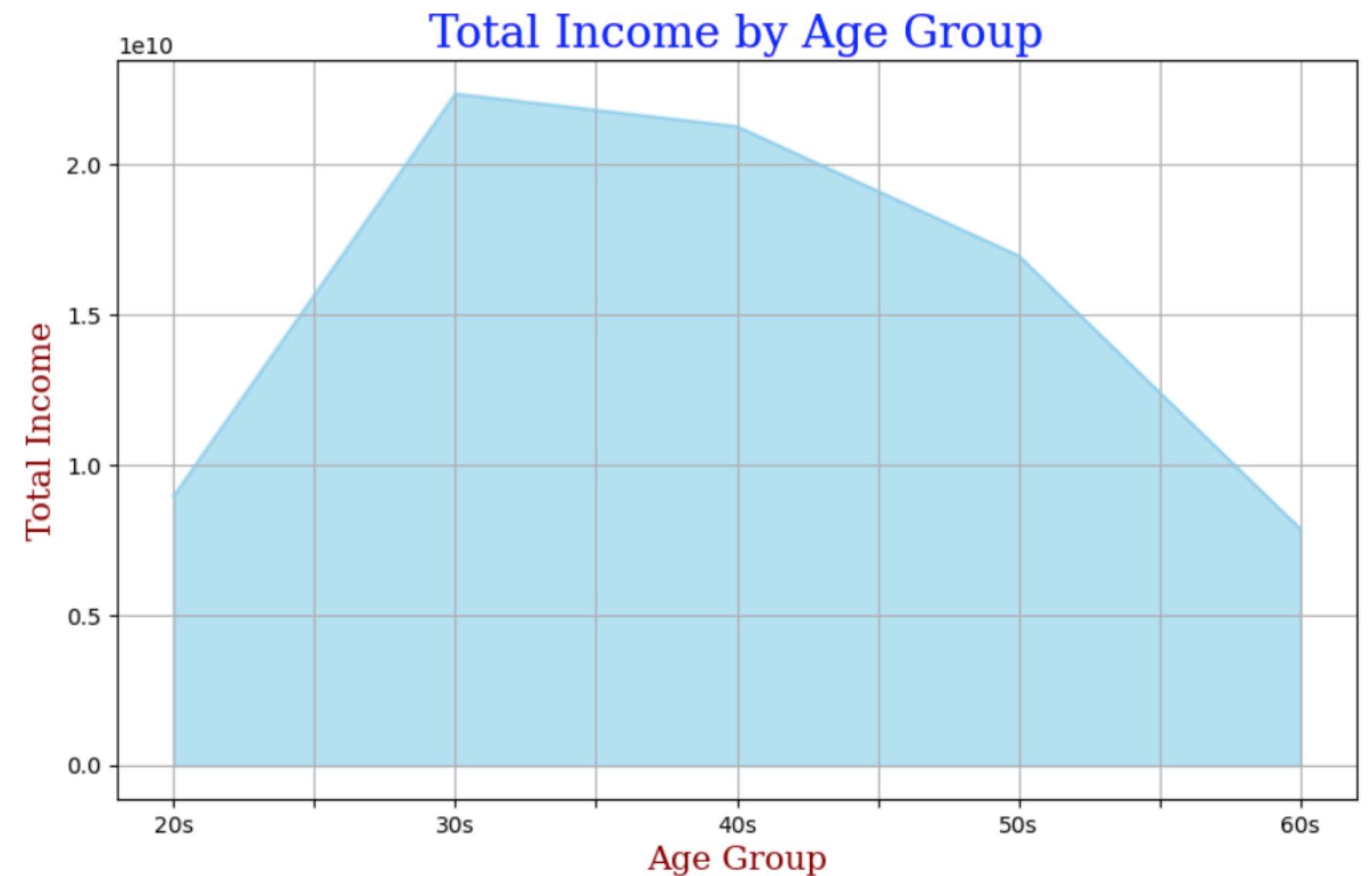
AGE and DAYS\_EMPLOYED show a negative correlation (-0.35). However, since DAYS\_EMPLOYED values are negative, this actually indicates that older individuals tend to have longer employment durations.



Most individuals have an income concentrated between 100,000 and 220,000

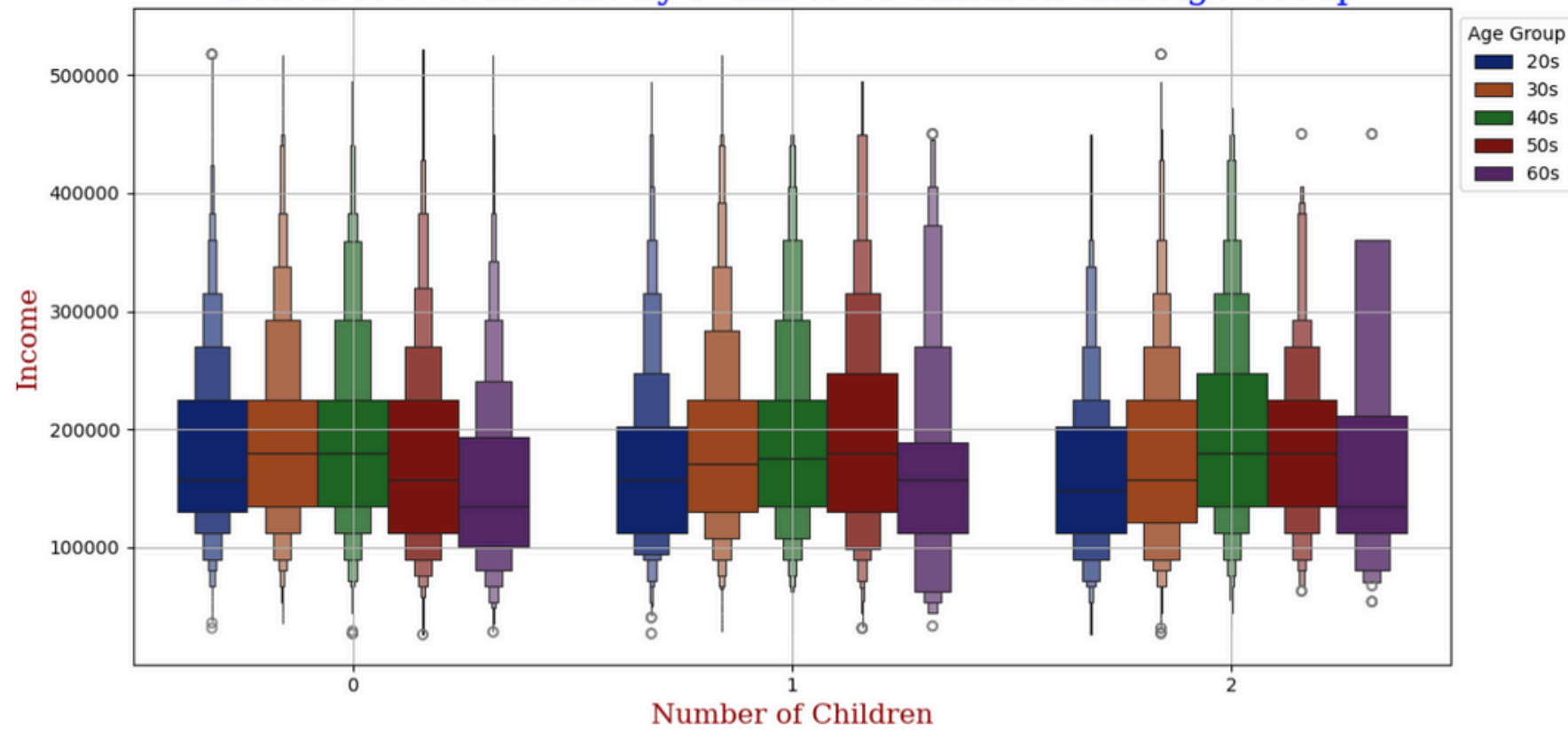


The regression line indicates a weak negative trend, suggesting that income slightly decreases with age



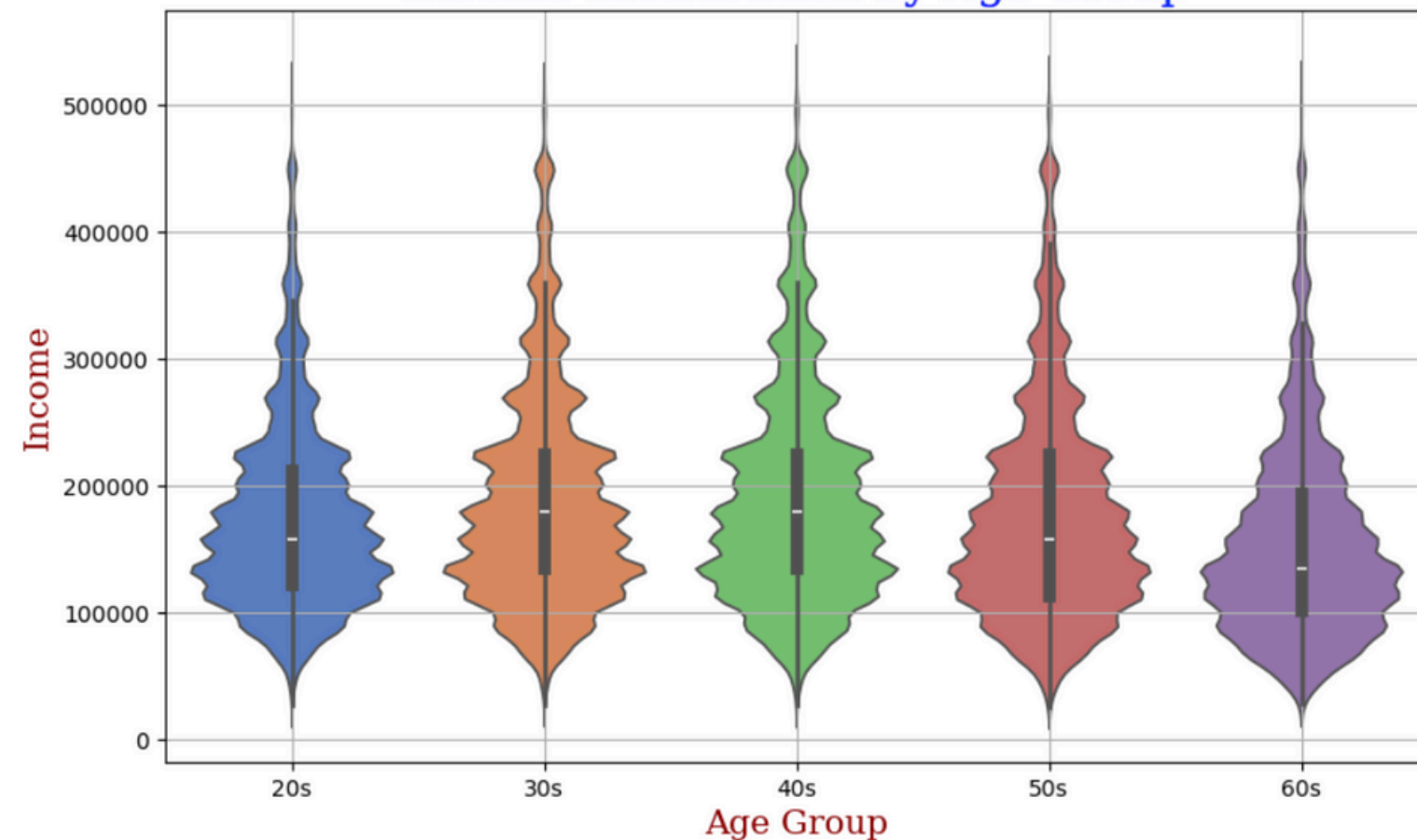
Income peaks in the 30s and declines gradually with age,

Boxen Plot of Income by Number of Children and Age Group



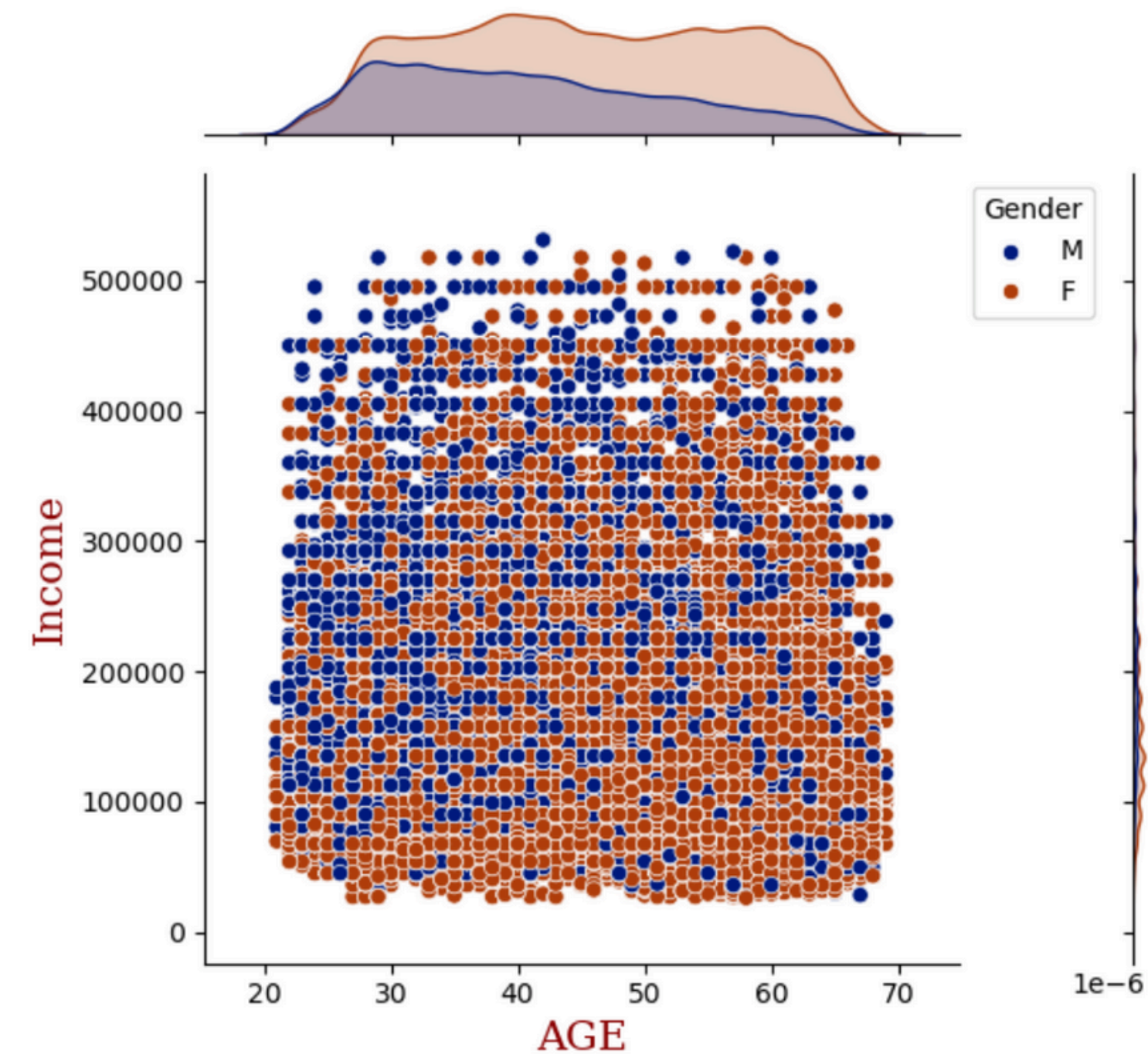
The median income is similar across child categories

Income Distribution by Age Group



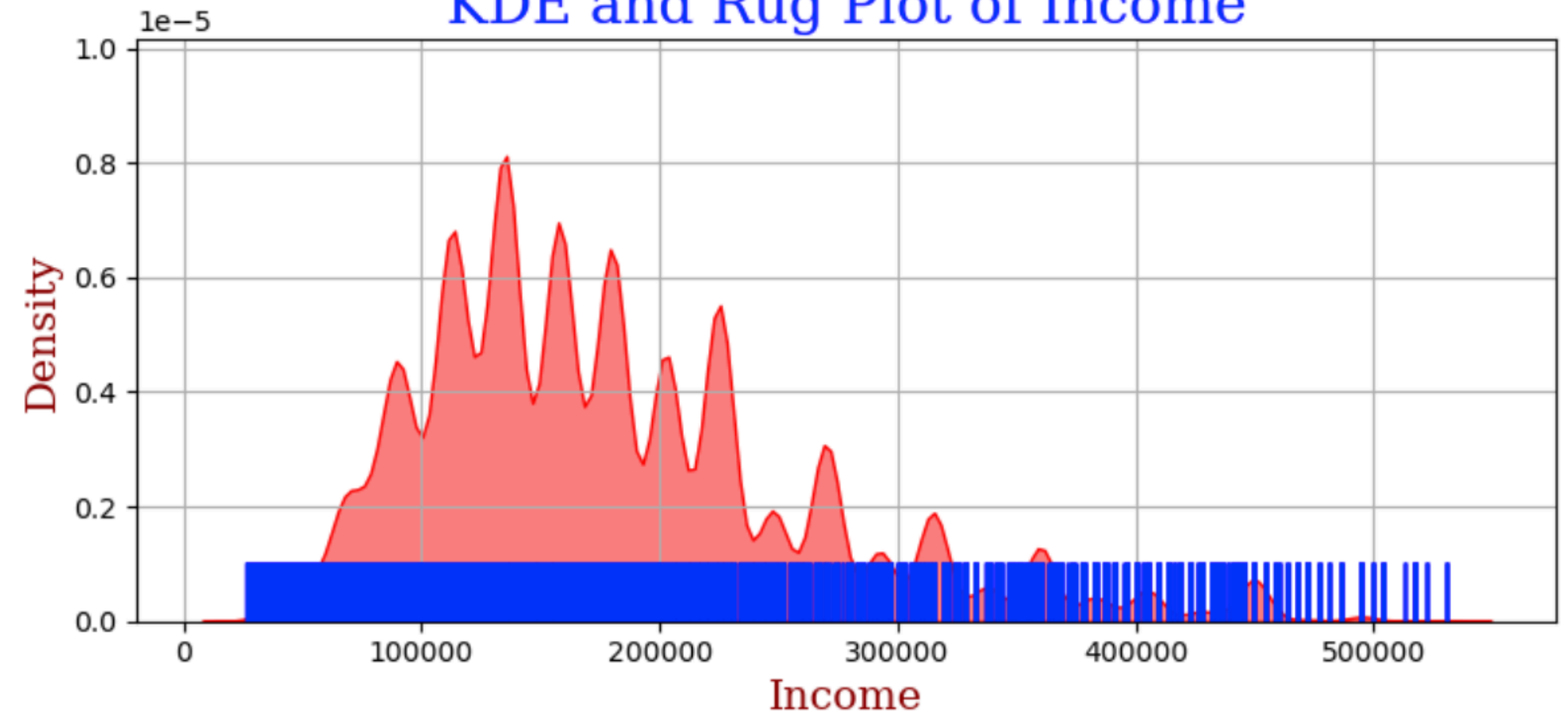
Most age groups exhibit a similar median income, but younger individuals have slightly higher variability in income levels.

Joint Plot of Age vs Income by Gender



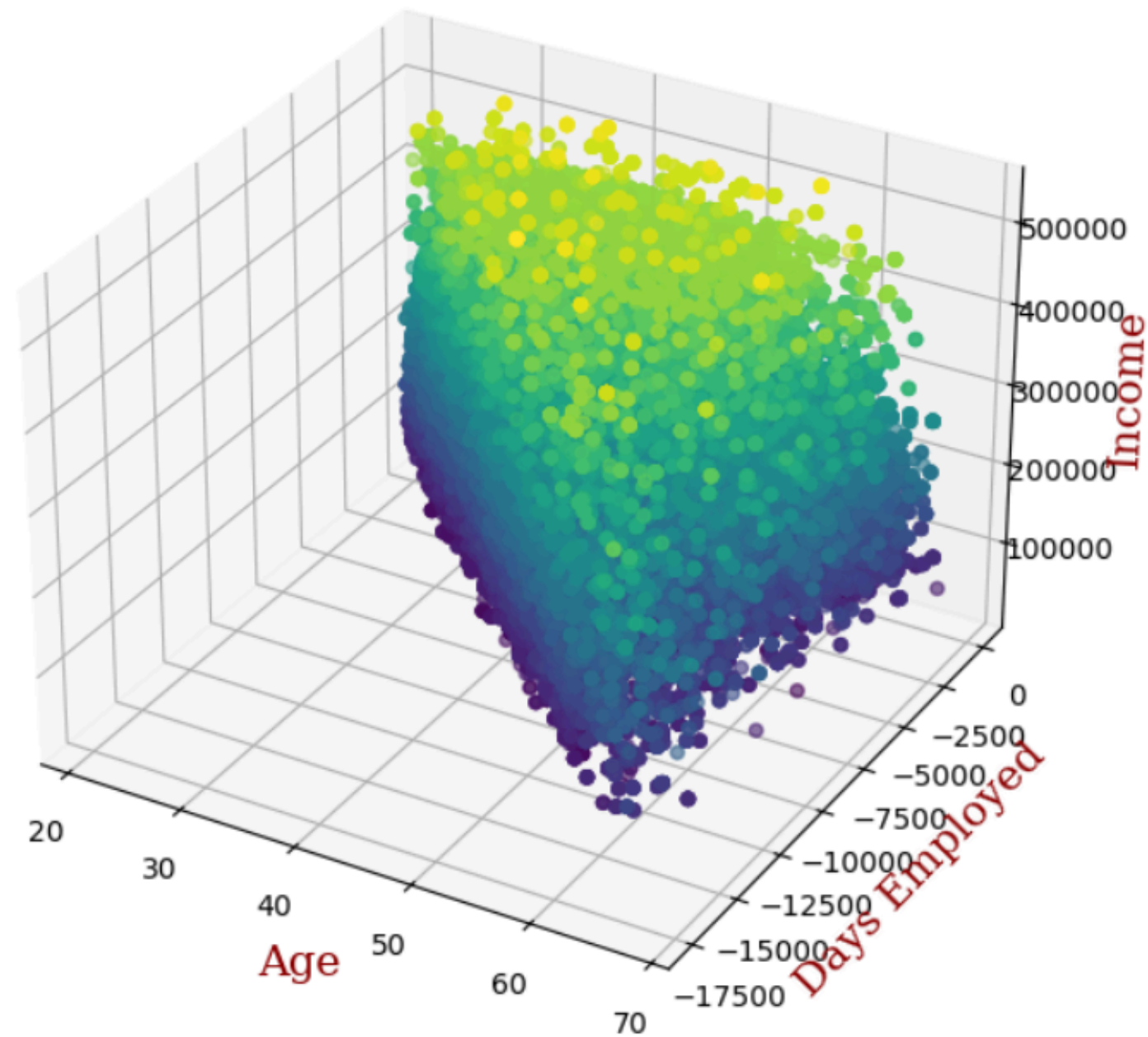
Income is spread similarly across both genders, with slight variations in density for different age ranges.

KDE and Rug Plot of Income



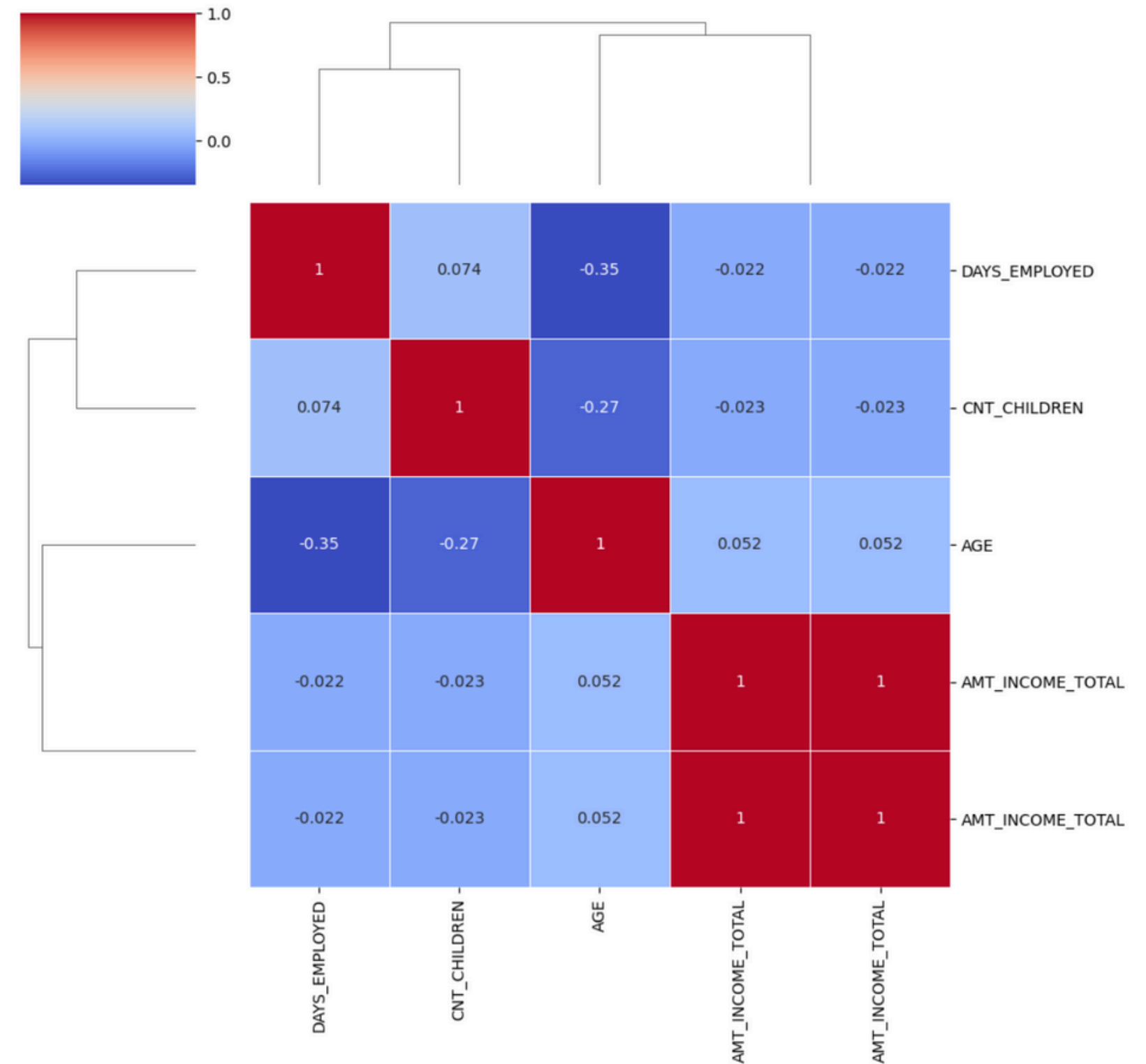
Income is concentrated below 300,000, with very few individuals earning above this threshold.

### 3D Scatter Plot

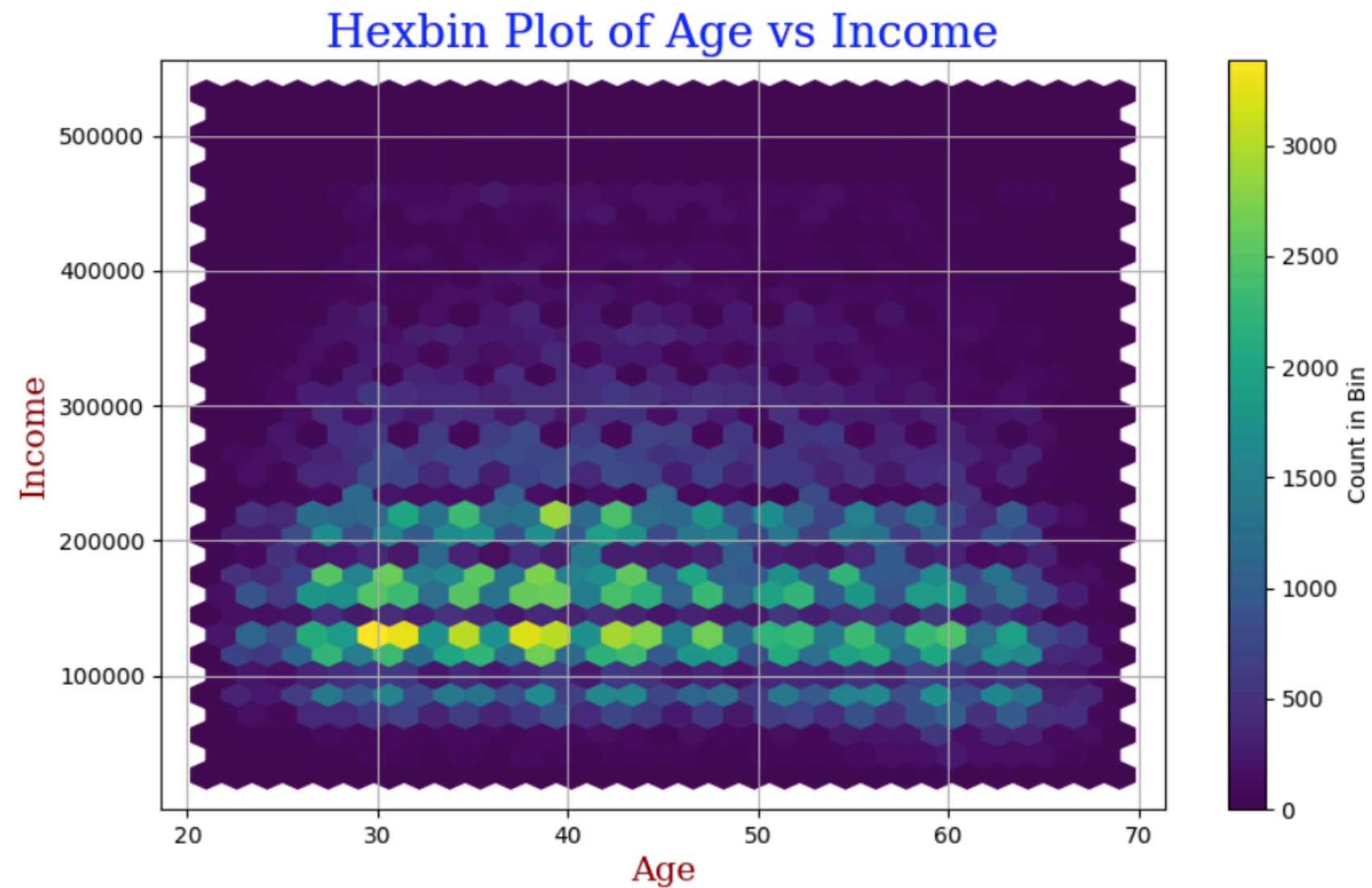


Income is mostly concentrated in the lower range, with high incomes being rare. Age and employment duration show little impact on income distribution.

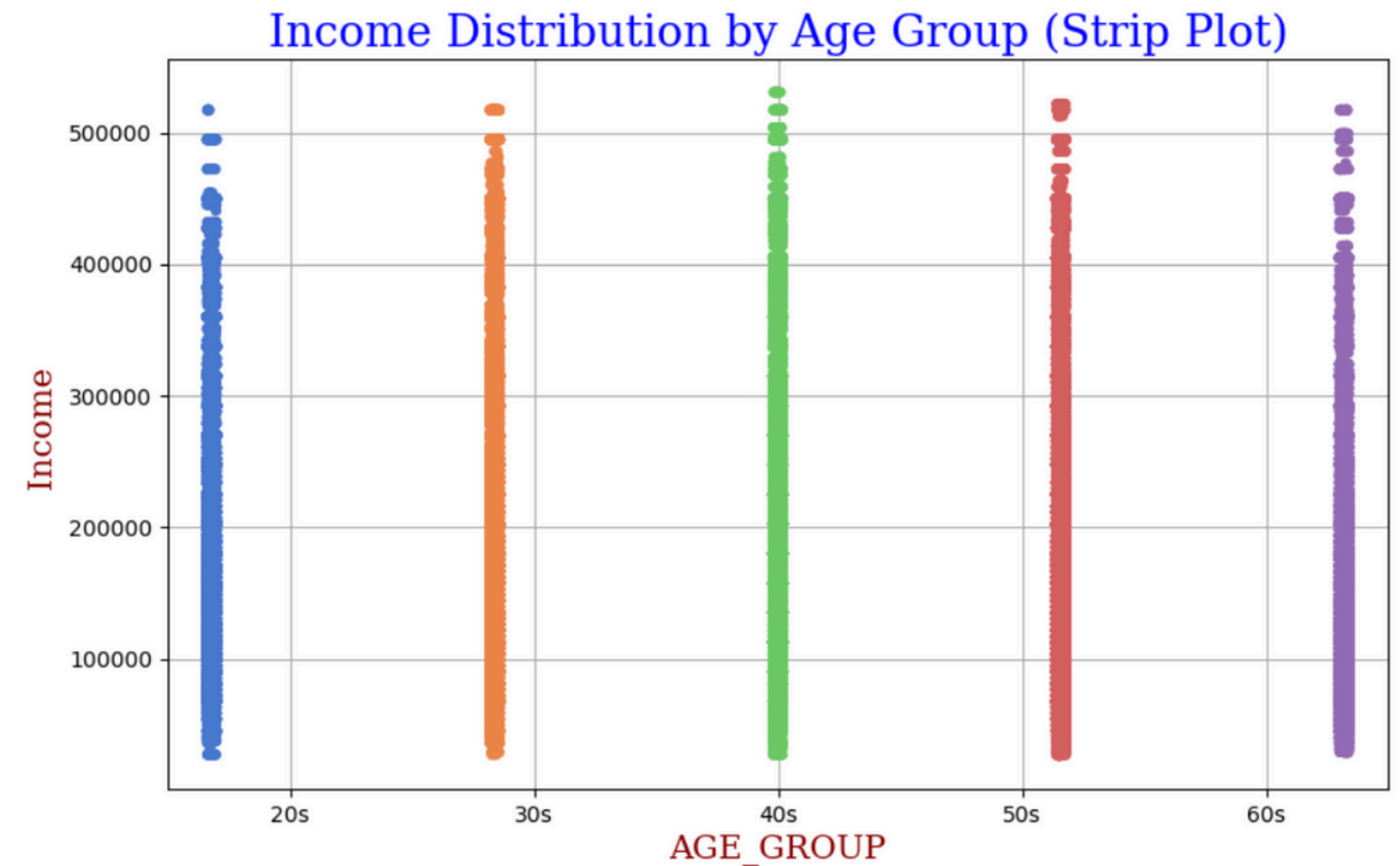
### Cluster Map of Selected Features



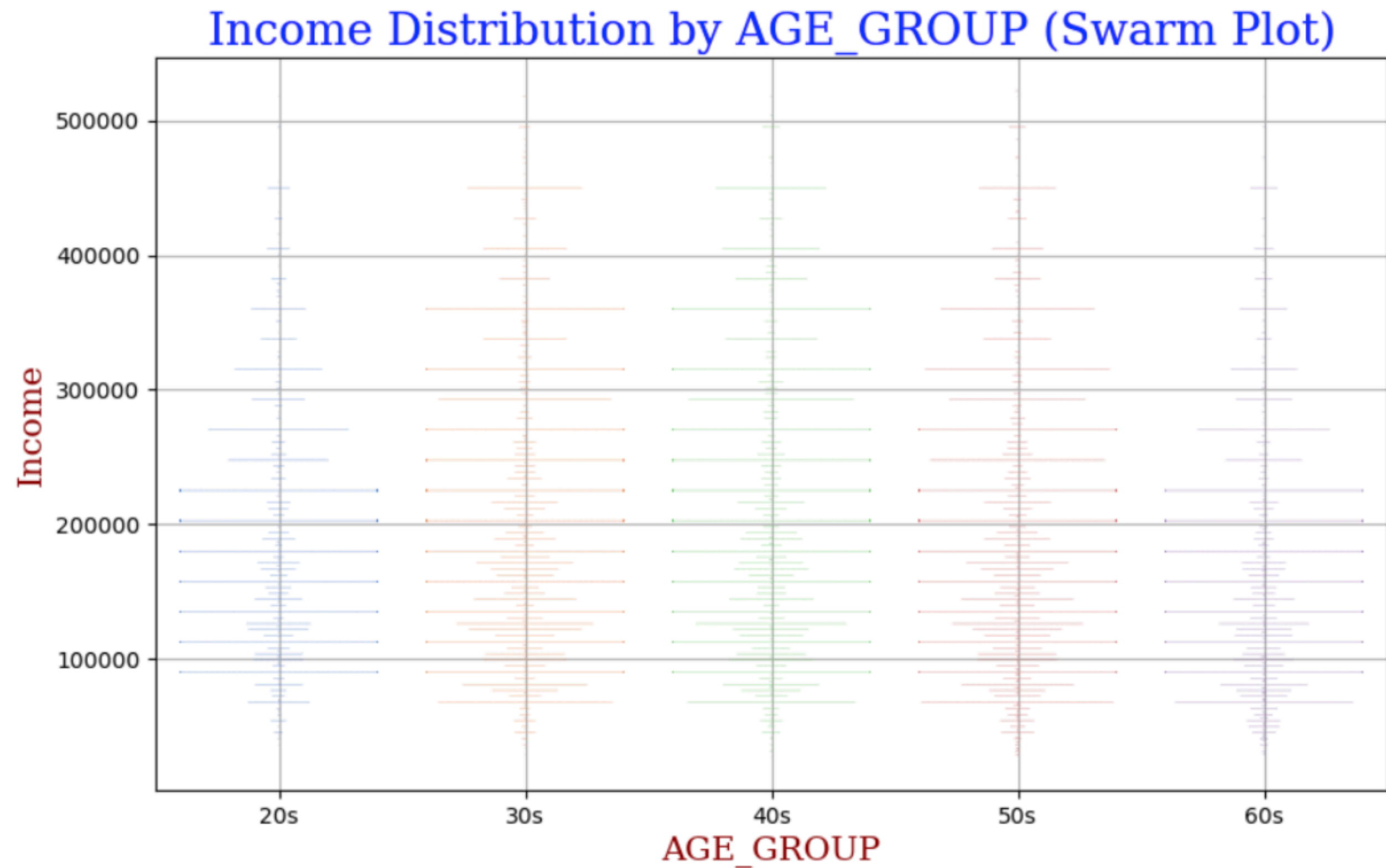
Strong correlations and clusters are indicated, such as the negative correlation between AGE and DAYS\_EMPLOYED.



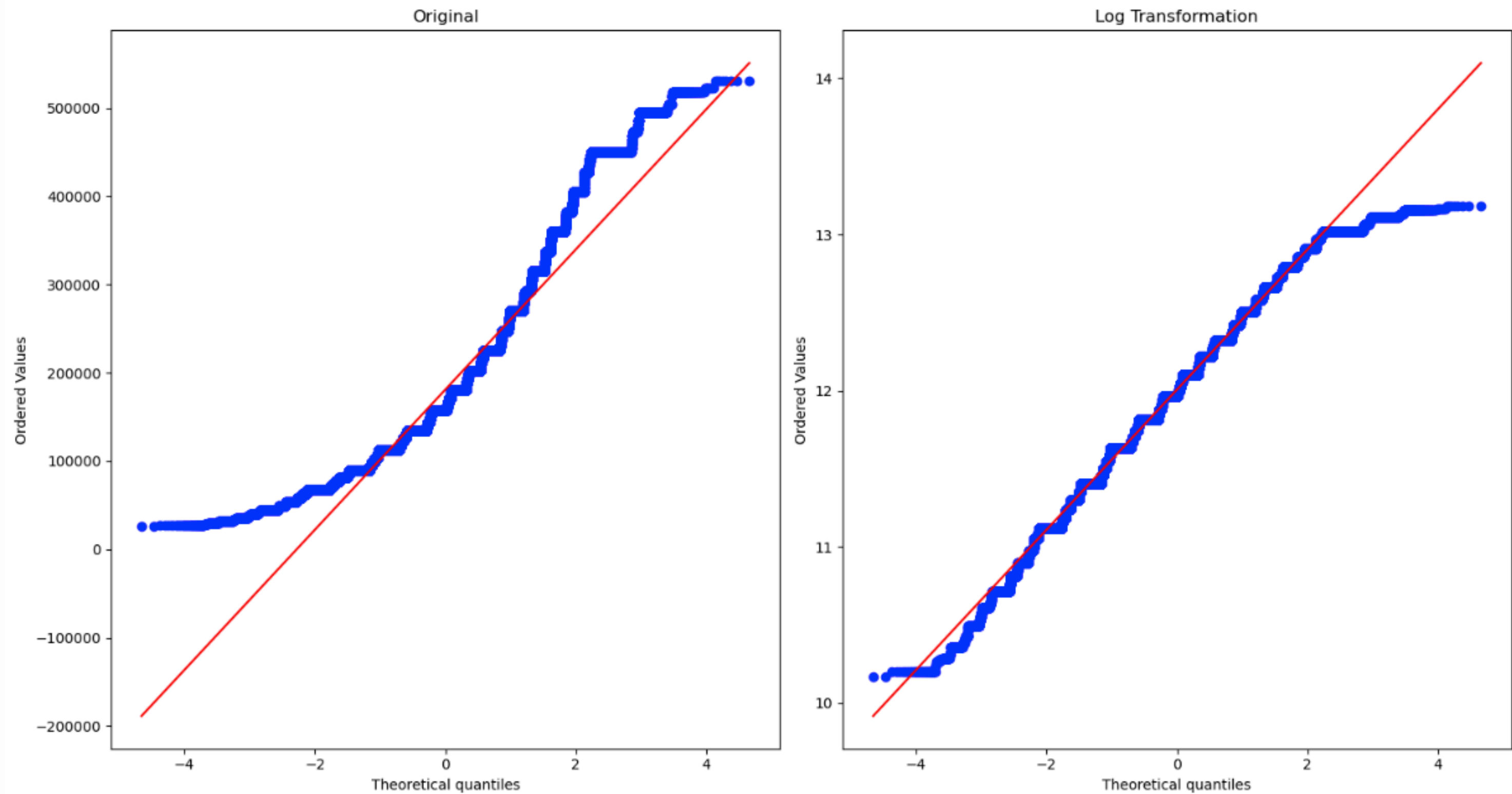
Most data points cluster around individuals with incomes between 100,000 and 300,000 across various ages, with denser regions represented by brighter colors.



The 30s and 40s exhibit slightly higher income levels compared to other groups, with a few outliers reaching up to 500,000

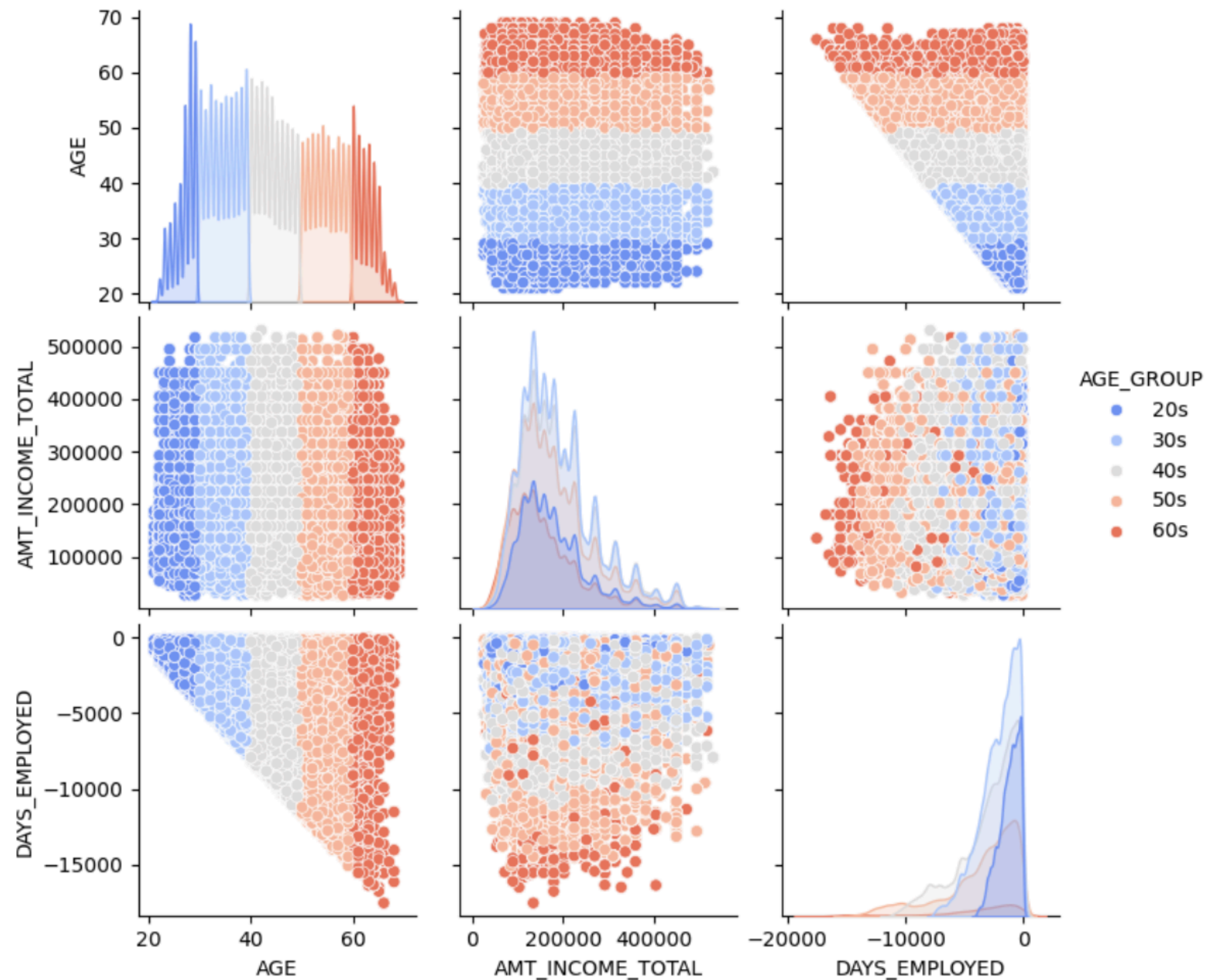


Each age group has a concentrated cluster of incomes around 200,000, with a few outliers in higher income brackets.



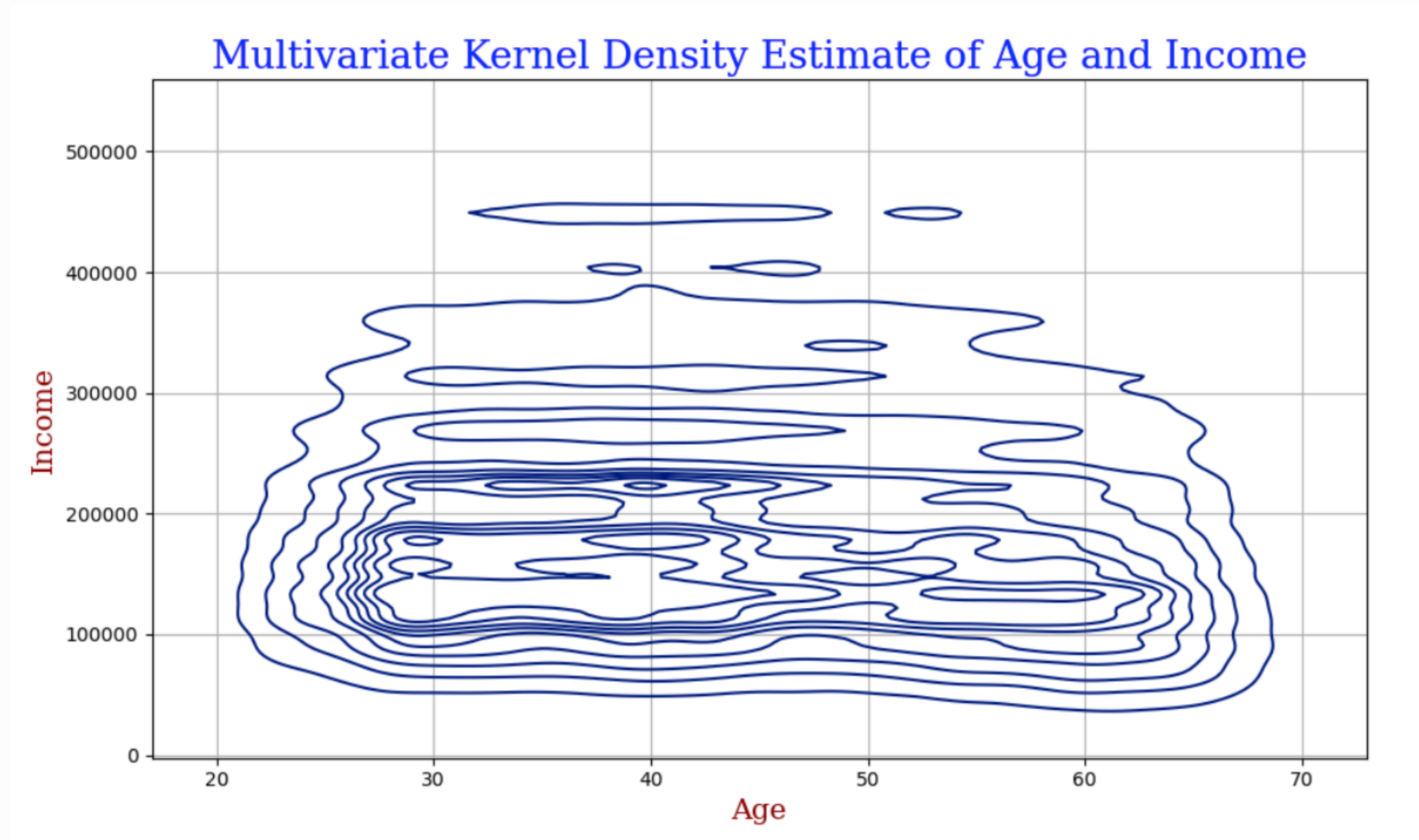
After applying the log transformation, the data became less skewed and more balanced, bringing it closer to a Gaussian distribution

Pair Plot

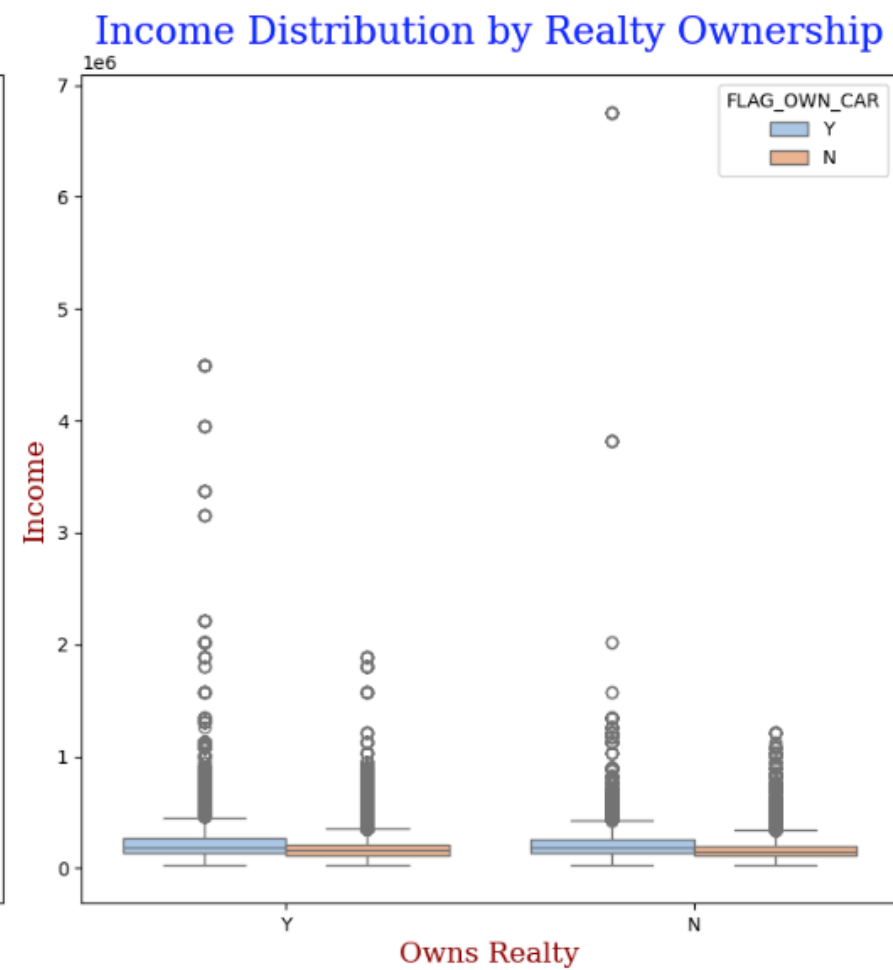
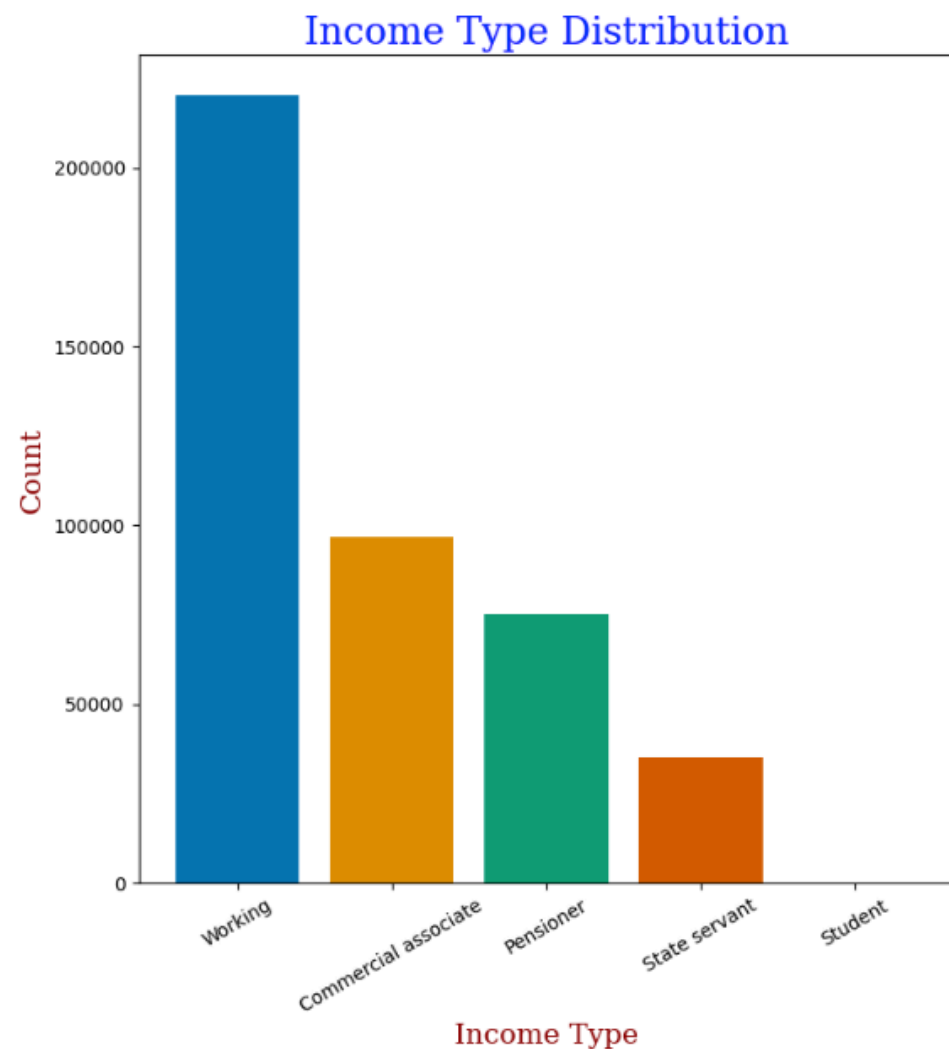
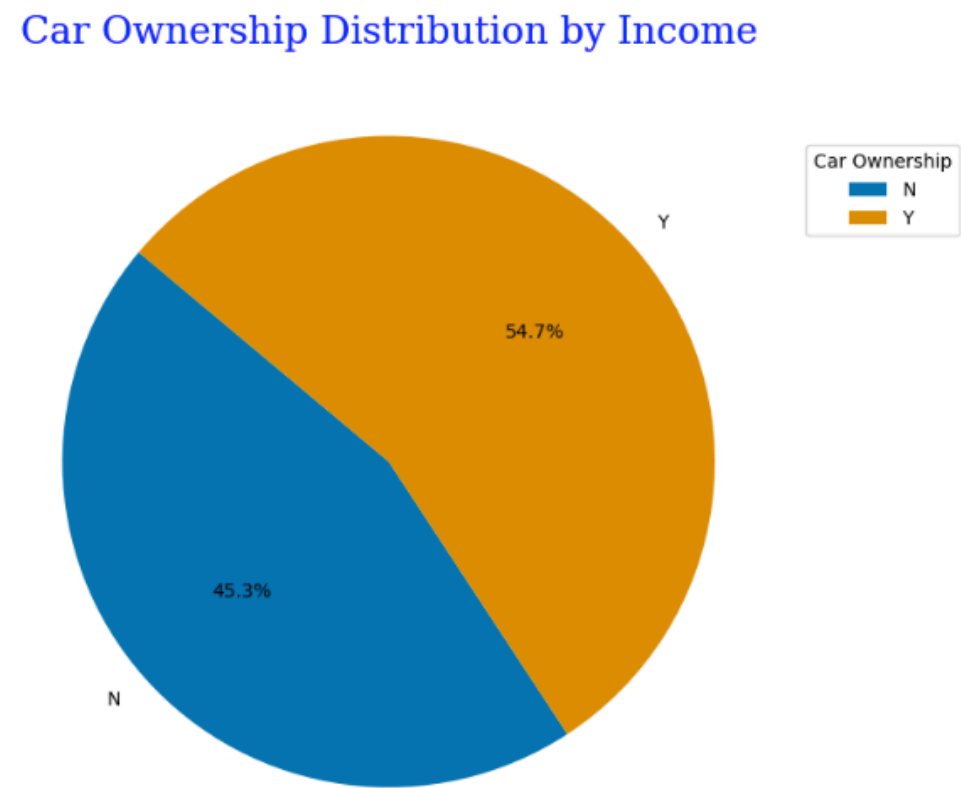
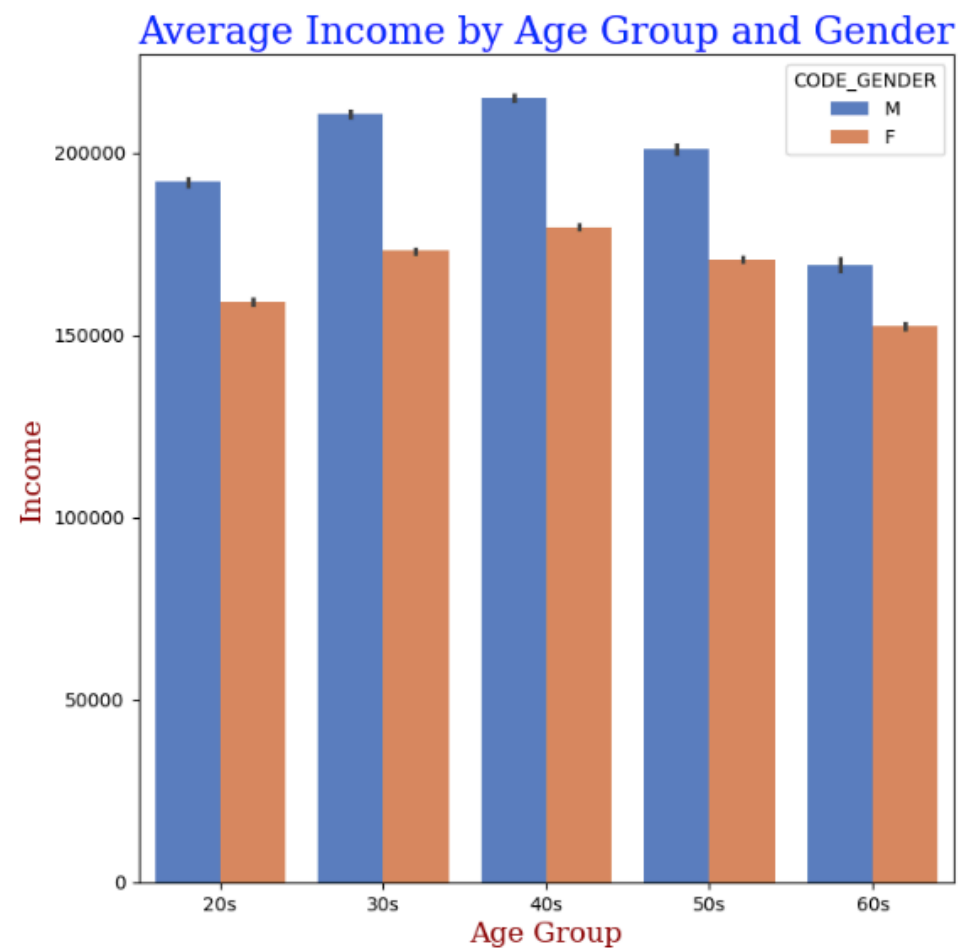


□ Income concentrated around 200,000 to 300,000

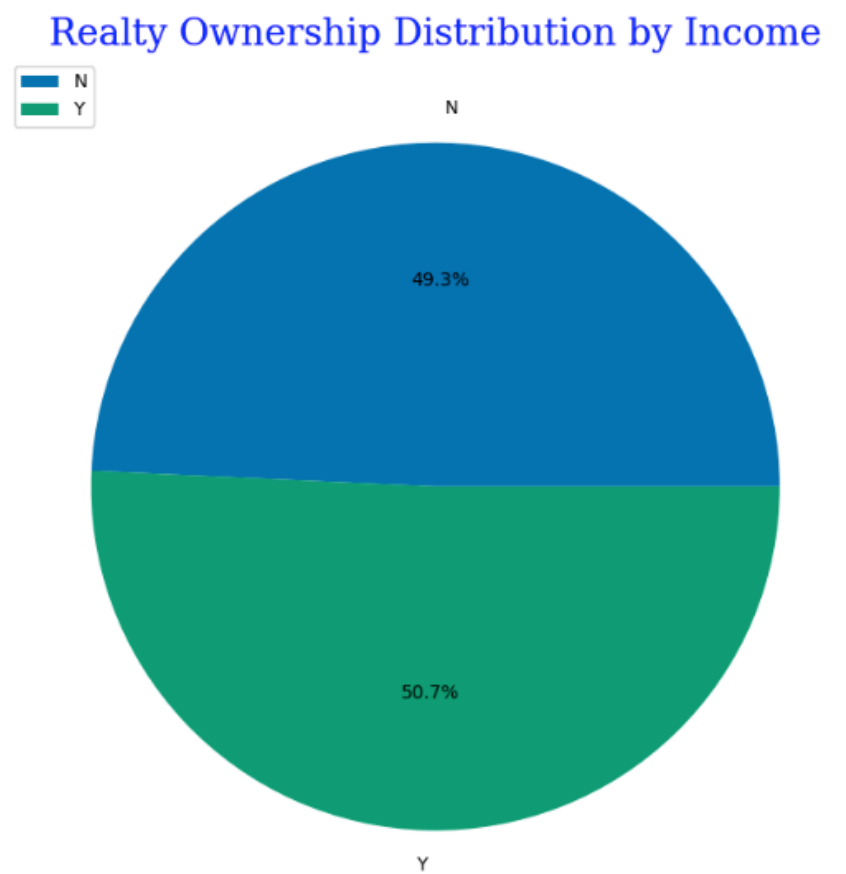
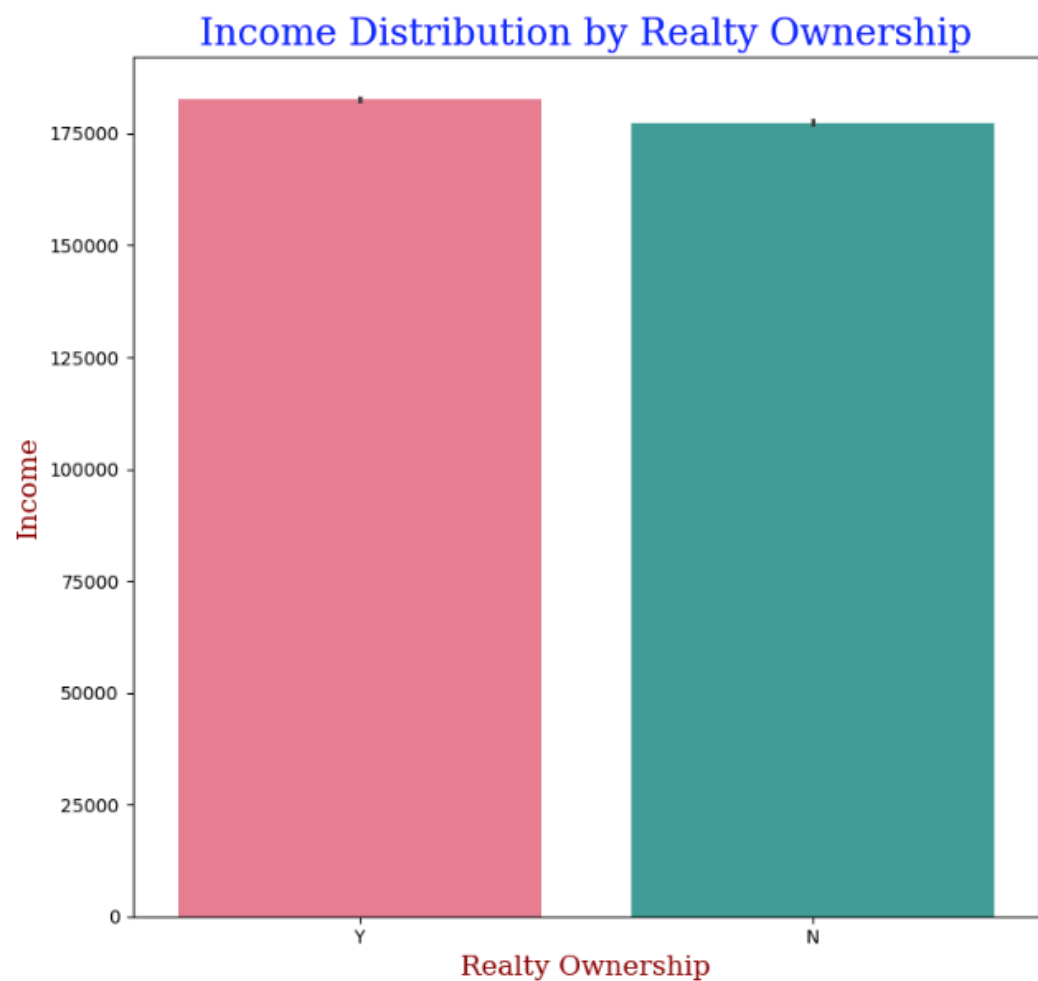
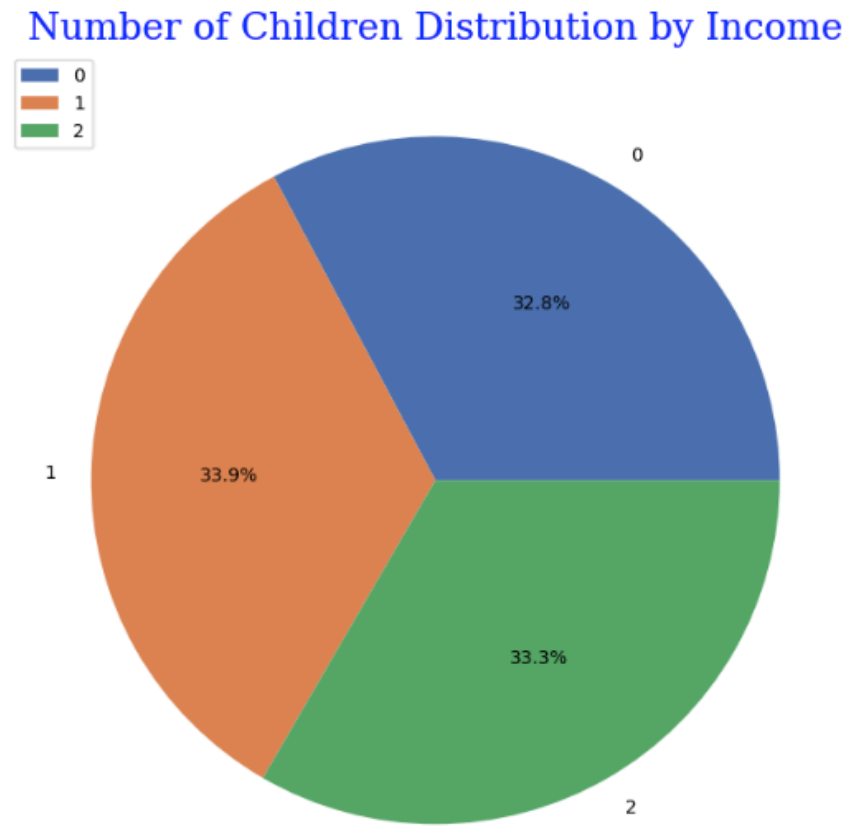
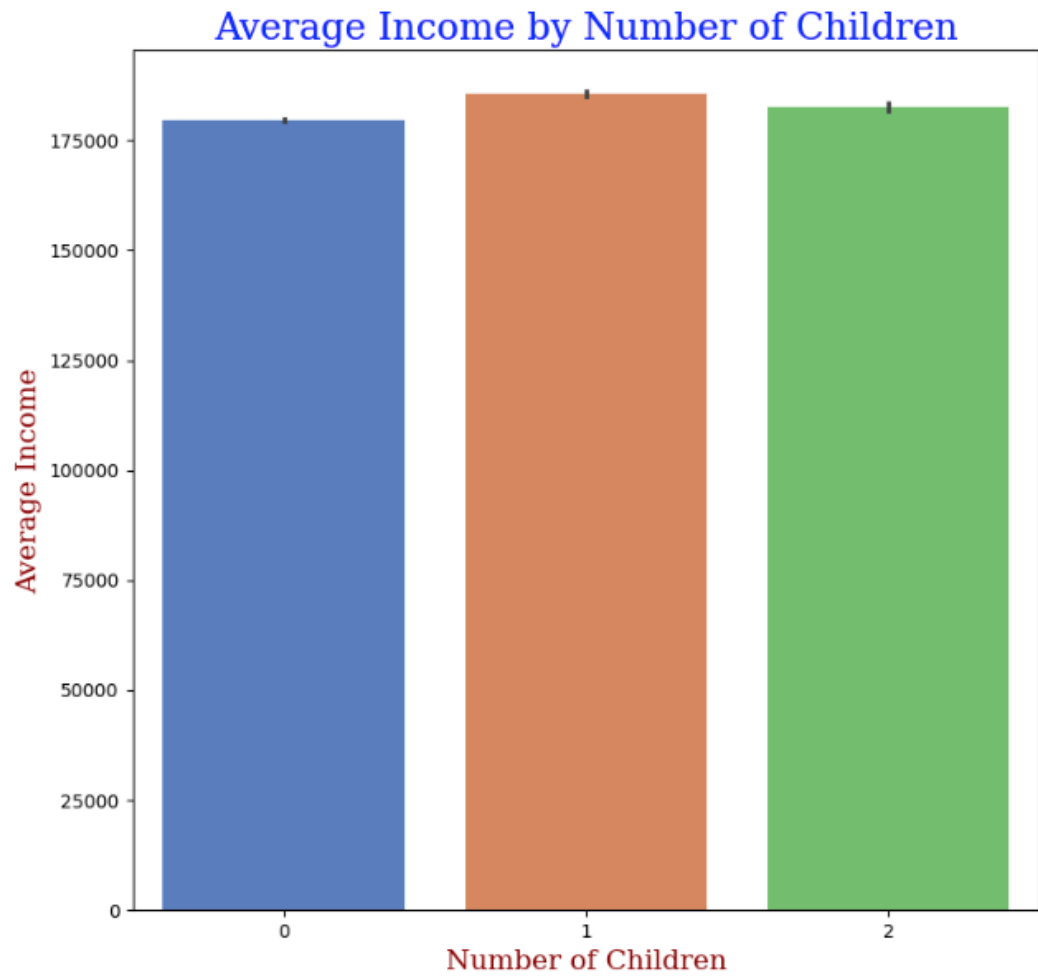
Clear positive correlation between age and days employed, as the employment days are represented with negative values in the dataset.



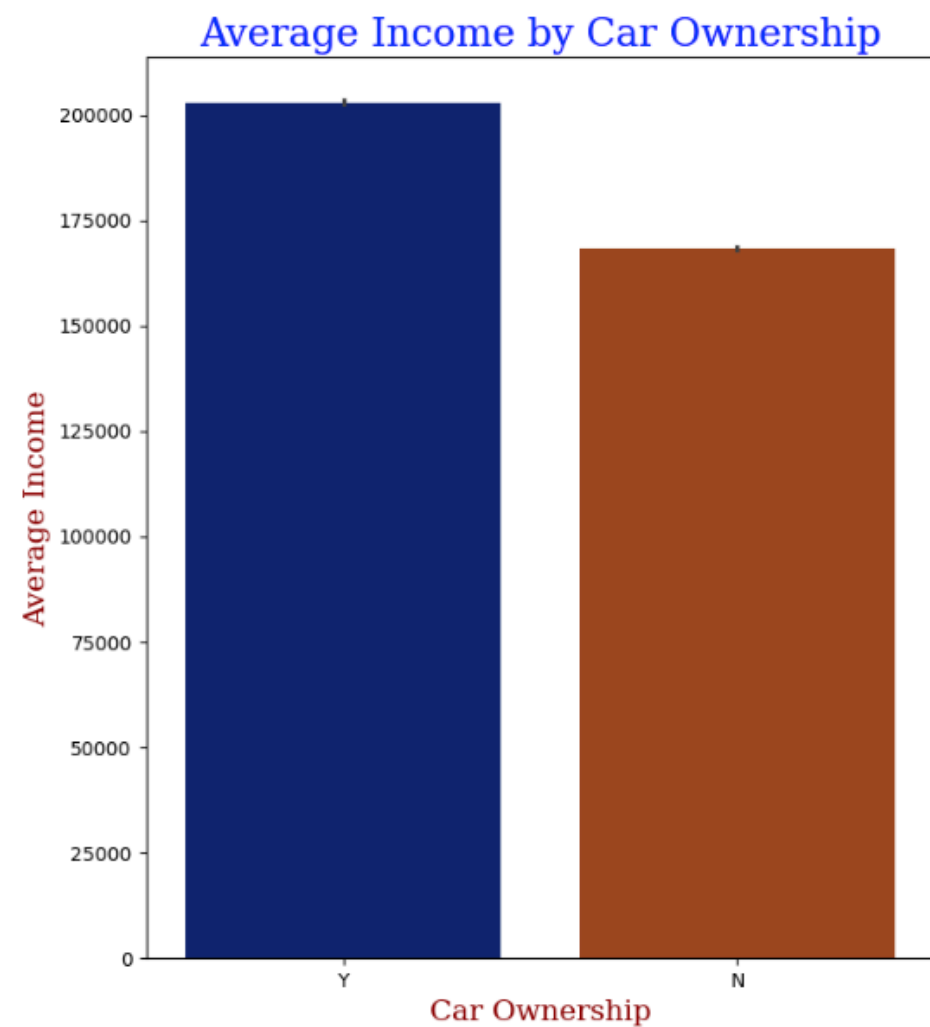
The area with the highest density, indicated by the closely packed contour lines, is observed in the age range of 30 to 40 years and the annual income range of \$100,000 to \$200,000.



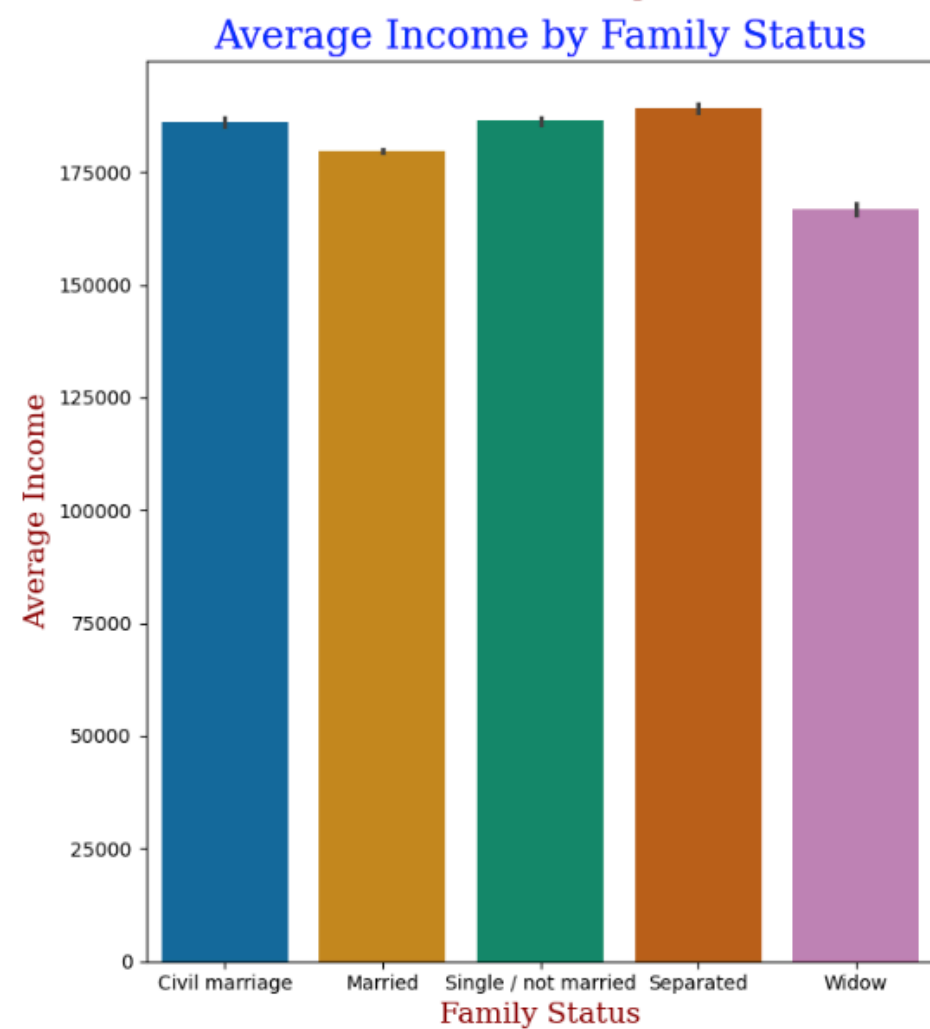
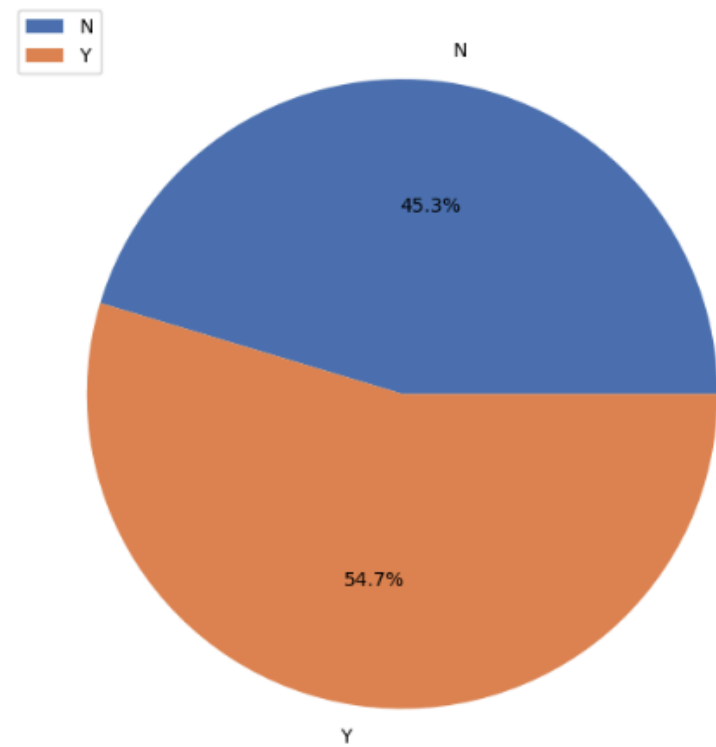
The dataset shows males earning more than females, with incomes peaking in their 40s. Most belong to the "Working" income type, car ownership is slightly more common, and real estate ownership has little impact on median income.



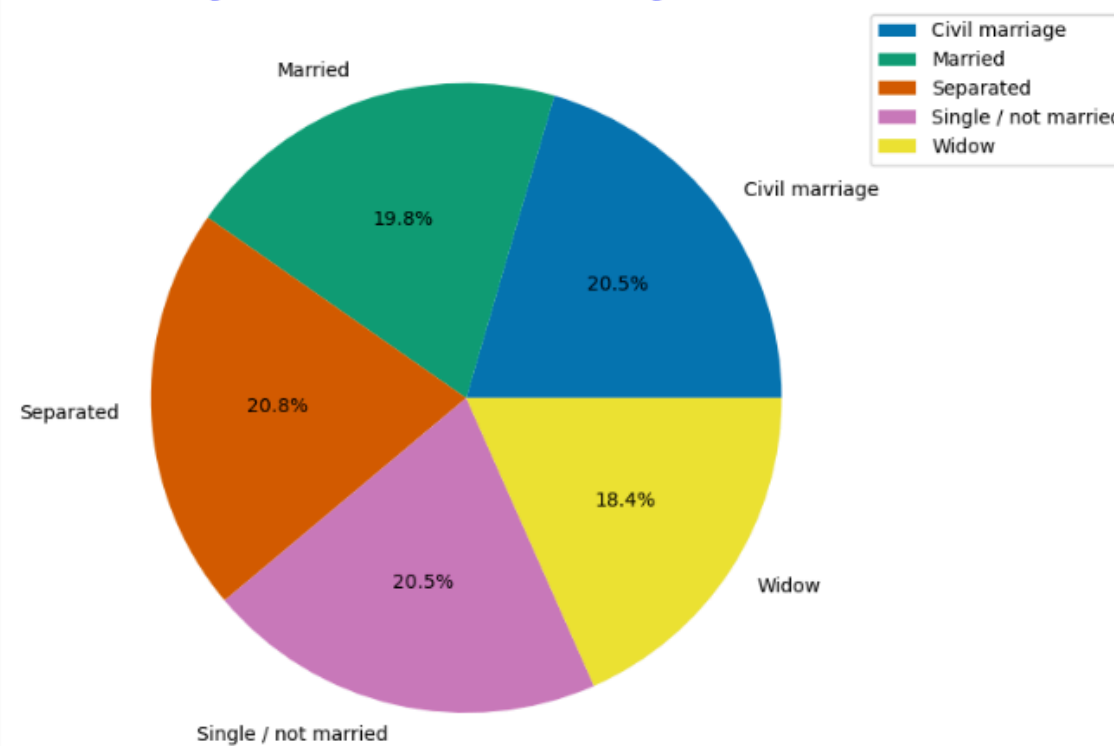
Income remains consistent regardless of the number of children or real estate ownership, showing no significant financial impact from either factor. The data suggests that these variables do not substantially influence income levels.



### Car Ownership Distribution by Income

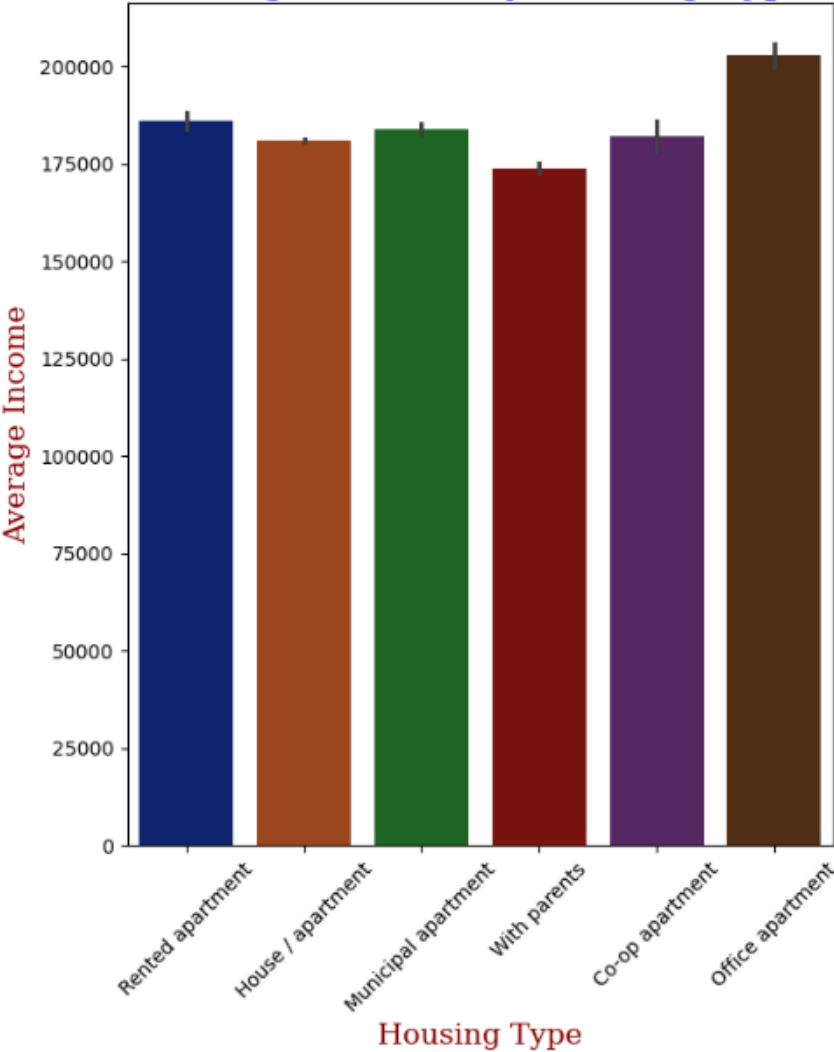


### Family Status Distribution by Income

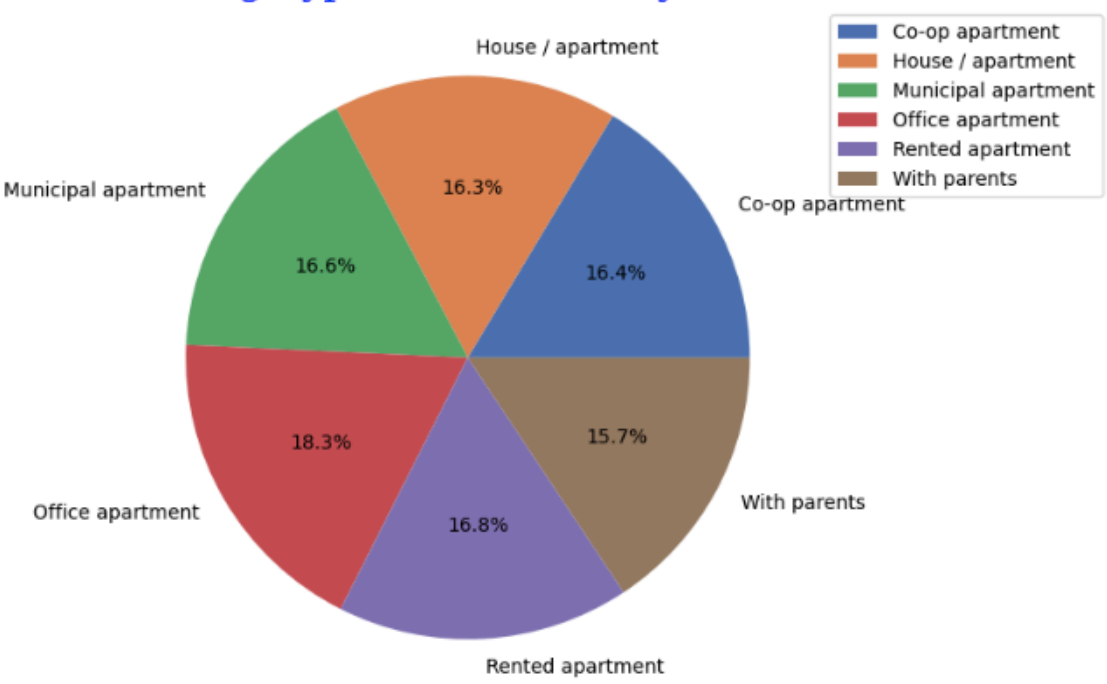


The plots reveal that car ownership is slightly more common at 54.7% and is associated with higher average incomes. In contrast, family status has less impact on income, though widows tend to earn the least.

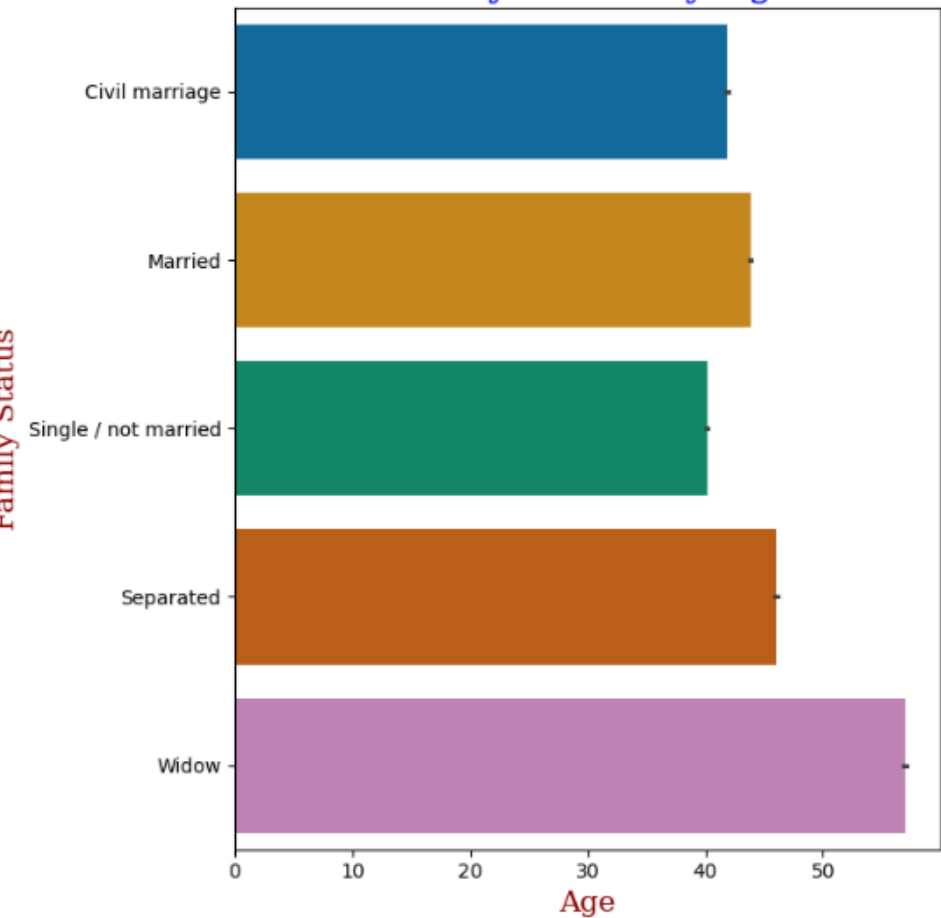
Average Income by Housing Type



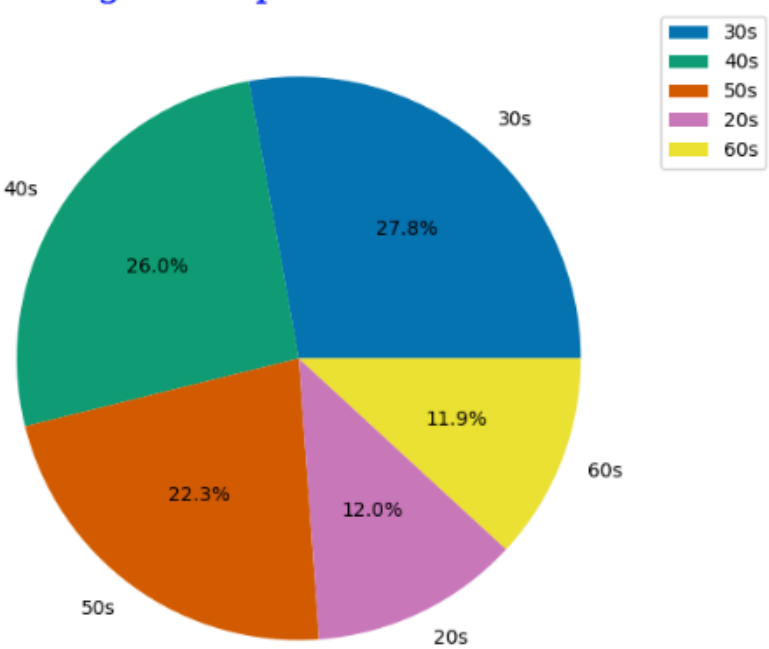
Housing Type Distribution by Income



Family Status by Age



Age Group Distribution



The charts reveal that average income is highest for individuals living in office apartments, with housing types evenly distributed across income groups. Family status shifts with age, as widows tend to be older, while the largest demographic is in their 30s.

PHASE 2

Data Type and Constraints

Categorical Columns

| Column ID          | Datatype                                    | Allow Null                          | Remove Duplicate         | Column ID           | Datatype                                    | Allow Null               | Remove Duplicate         |
|--------------------|---------------------------------------------|-------------------------------------|--------------------------|---------------------|---------------------------------------------|--------------------------|--------------------------|
| ID                 | <div>String</div> <div>×</div> <div>▼</div> | <input type="checkbox"/>            | <input type="checkbox"/> | CODE_GENDER         | <div>String</div> <div>×</div> <div>▼</div> | <input type="checkbox"/> | <input type="checkbox"/> |
| FLAG_OWN_CAR       | <div>String</div> <div>×</div> <div>▼</div> | <input type="checkbox"/>            | <input type="checkbox"/> | FLAG_OWN_REALTY     | <div>String</div> <div>×</div> <div>▼</div> | <input type="checkbox"/> | <input type="checkbox"/> |
| NAME_INCOME_TYPE   | <div>String</div> <div>×</div> <div>▼</div> | <input type="checkbox"/>            | <input type="checkbox"/> | NAME_EDUCATION_TYPE | <div>String</div> <div>×</div> <div>▼</div> | <input type="checkbox"/> | <input type="checkbox"/> |
| NAME_FAMILY_STATUS | <div>String</div> <div>×</div> <div>▼</div> | <input type="checkbox"/>            | <input type="checkbox"/> | NAME_HOUSING_TYPE   | <div>String</div> <div>×</div> <div>▼</div> | <input type="checkbox"/> | <input type="checkbox"/> |
| FLAG_MOBIL         | <div>String</div> <div>×</div> <div>▼</div> | <input type="checkbox"/>            | <input type="checkbox"/> | FLAG_WORK_PHONE     | <div>String</div> <div>×</div> <div>▼</div> | <input type="checkbox"/> | <input type="checkbox"/> |
| FLAG_PHONE         | <div>String</div> <div>×</div> <div>▼</div> | <input type="checkbox"/>            | <input type="checkbox"/> | FLAG_EMAIL          | <div>String</div> <div>×</div> <div>▼</div> | <input type="checkbox"/> | <input type="checkbox"/> |
| OCCUPATION_TYPE    | <div>String</div> <div>×</div> <div>▼</div> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | AGE_GROUP           | <div>String</div> <div>×</div> <div>▼</div> | <input type="checkbox"/> | <input type="checkbox"/> |

Numerical Columns

| Column ID       | Datatype                                     | Allow Null               | Remove Duplicate         | Column ID        | Datatype                                     | Allow Null               | Remove Duplicate         |
|-----------------|----------------------------------------------|--------------------------|--------------------------|------------------|----------------------------------------------|--------------------------|--------------------------|
| CNT_CHILDREN    | <div>Integer</div> <div>×</div> <div>▼</div> | <input type="checkbox"/> | <input type="checkbox"/> | AMT_INCOME_TOTAL | <div>Float</div> <div>×</div> <div>▼</div>   | <input type="checkbox"/> | <input type="checkbox"/> |
| DAYS_BIRTH      | <div>Integer</div> <div>×</div> <div>▼</div> | <input type="checkbox"/> | <input type="checkbox"/> | DAYS_EMPLOYED    | <div>Integer</div> <div>×</div> <div>▼</div> | <input type="checkbox"/> | <input type="checkbox"/> |
| CNT_FAM_MEMBERS | <div>Float</div> <div>×</div> <div>▼</div>   | <input type="checkbox"/> | <input type="checkbox"/> | AGE              | <div>Float</div> <div>×</div> <div>▼</div>   | <input type="checkbox"/> | <input type="checkbox"/> |

Submit

DataFrame Table

| ID      | CODE_GENDER | FLAG_OWN_CAR | FLAG_OWN_REALTY | CNT_CHILDREN | AMT_INCOME_TOTAL | NAME_INCOME_TYPE     | NAME_EDUCATION_TY             | NAME_FAMILY_STATUS   | NAME_HOUSING_TYPE | DAYS_BIRTH | DAYS_EMPLOYED | FLAG_MOBIL | FLAG_WORK_PHONE | FLAG_PHONE | FLAG_EMAIL | OCCUPATION_TYPE | CNT_FAM_ME |
|---------|-------------|--------------|-----------------|--------------|------------------|----------------------|-------------------------------|----------------------|-------------------|------------|---------------|------------|-----------------|------------|------------|-----------------|------------|
| 5008804 | M           | Y            | Y               | 0            | 427500           | Working              | Higher education              | Civil marriage       | Rented apartment  | −12005     | −4542         | 1          | 1               | 0          | 0          |                 |            |
| 5008805 | M           | Y            | Y               | 0            | 427500           | Working              | Higher education              | Civil marriage       | Rented apartment  | −12005     | −4542         | 1          | 1               | 0          | 0          |                 |            |
| 5008806 | M           | Y            | Y               | 0            | 112500           | Working              | Secondary / secondary special | Married              | House / apartment | −21474     | −1134         | 1          | 0               | 0          | 0          | Security staff  |            |
| 5008808 | F           | N            | Y               | 0            | 270000           | Commercial associate | Secondary / secondary special | Single / not married | House / apartment | −19110     | −3051         | 1          | 0               | 1          | 1          | Sales staff     |            |
| 5008809 | F           | N            | Y               | 0            | 270000           | Commercial associate | Secondary / secondary special | Single / not married | House / apartment | −19110     | −3051         | 1          | 0               | 1          | 1          | Sales staff     |            |
| 5008810 | F           | N            | Y               | 0            | 270000           | Commercial associate | Secondary / secondary special | Single / not married | House / apartment | −19110     | −3051         | 1          | 0               | 1          | 1          | Sales staff     |            |

Numerical Columns

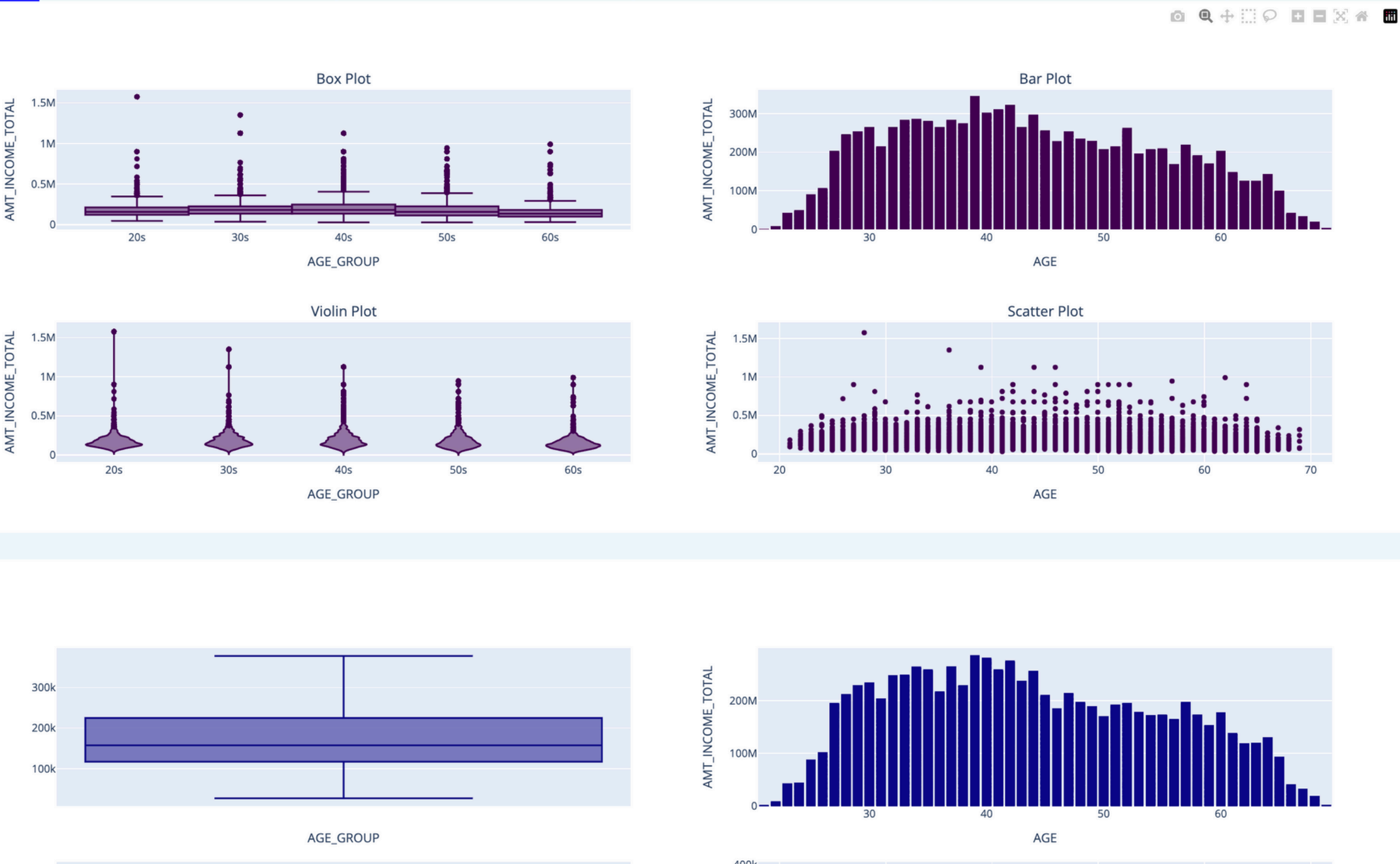
Choose a column to plot

- CNT\_CHILDREN
- DAYS\_BIRTH
- CNT\_FAM\_MEMBERS
- AMT\_INCOME\_TOTAL
- DAYS\_EMPLOYED
- AGE

AMT\_INCOME\_TOTAL

Choose a method to detect outliers

- IQR
- Z-score



Numerical Columns

Choose columns to PCA

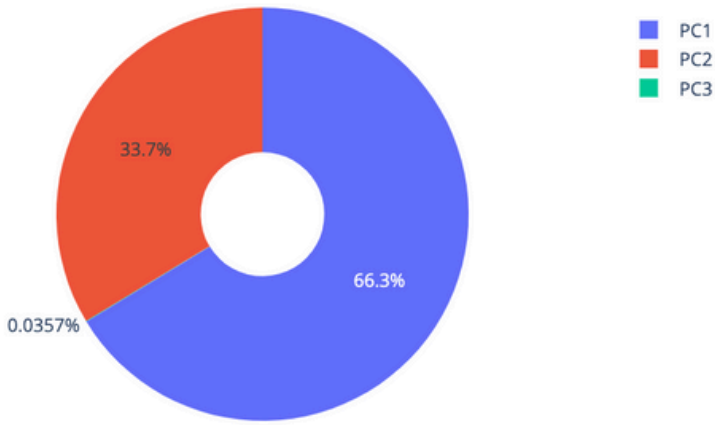
- ☒ CNT\_CHILDREN
- ☒ AMT\_INCOME\_TOTAL
- ☒ DAYS\_BIRTH
- ☒ DAYS\_EMPLOYED
- ☐ CNT\_FAM\_MEMBERS
- ☐ AGE

Dimensionality setting : 

Submit

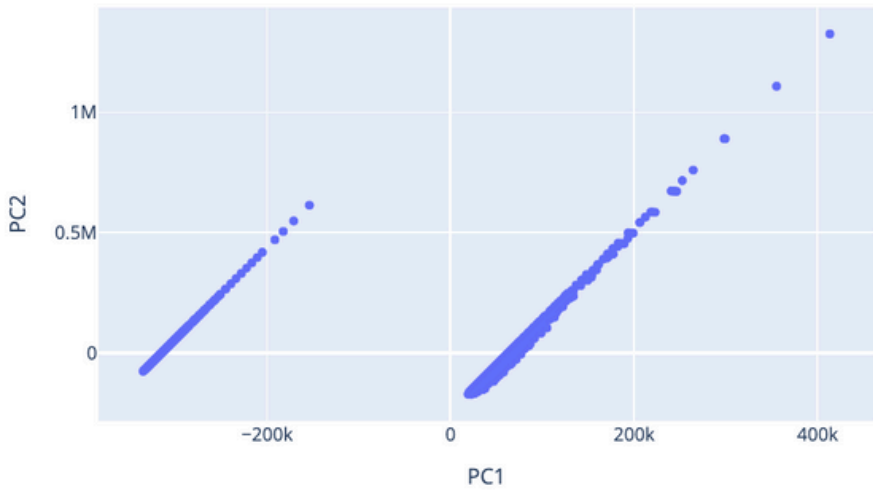
Pie plot of PCA explained variance ratio

PCA Explained Variance



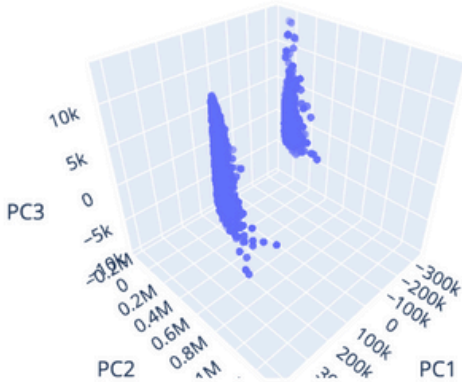
2D plot when dimension is 2

PCA 2D



3D plot when dimension is 3

PCA 3D



Numerical Columns

Choose columns to PCA

☐ CNT\_CHILDREN

☐ DAYS\_EMPLOYED

☐ AMT\_INCOME\_TOTAL

☐ CNT\_FAM\_MEMBERS

☒ DAYS\_BIRTH

☐ AGE

Sampling size :

100110021003100410051006100710081009100

Choose one:

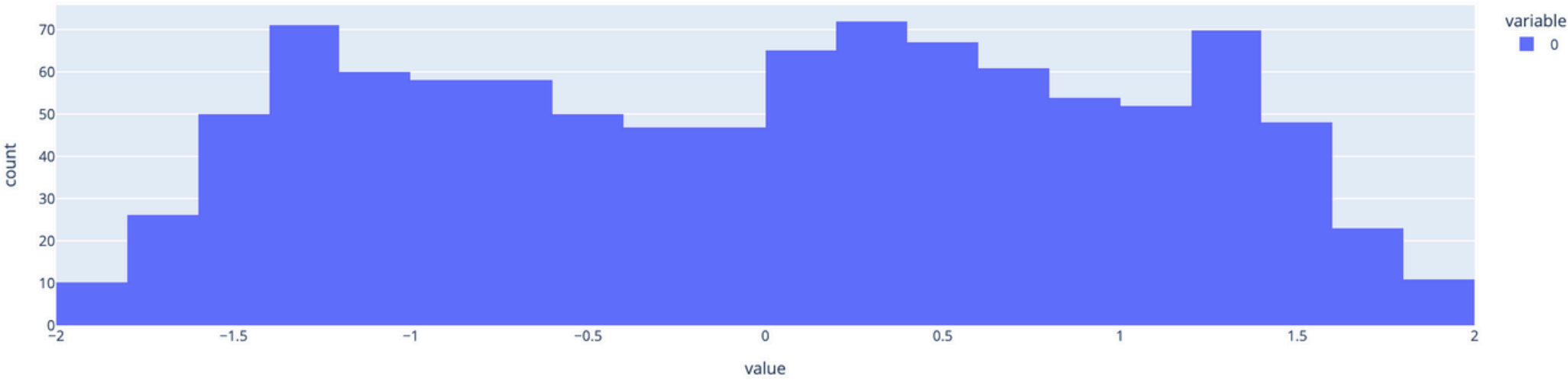
☒ Kolmogorov-Smirnov

☐ Shapiro-Wilk

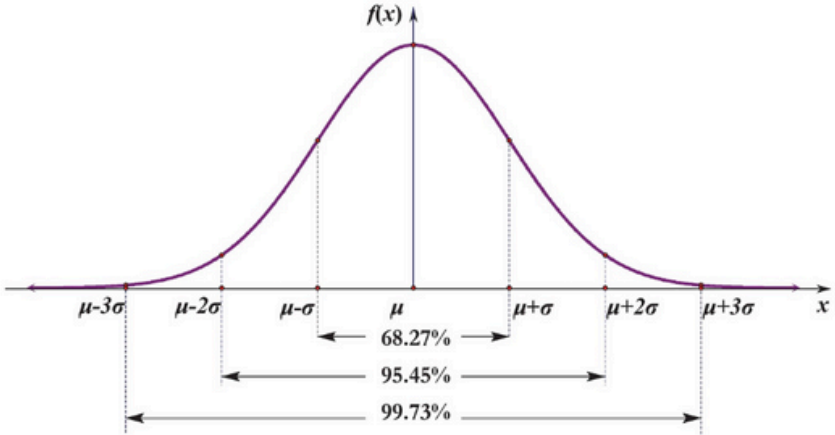
☐ D'Agostino-Pearson

Submit

Kolmogorov-Smirnov test, p-value: 0.00



Reference Img of Normal Distribution :



Choose a column to transform (Numerical) :

AMT\_INCOME\_TOTAL

×

▼

Choose a transformation method :

☐ Normalization

☐ Standardization

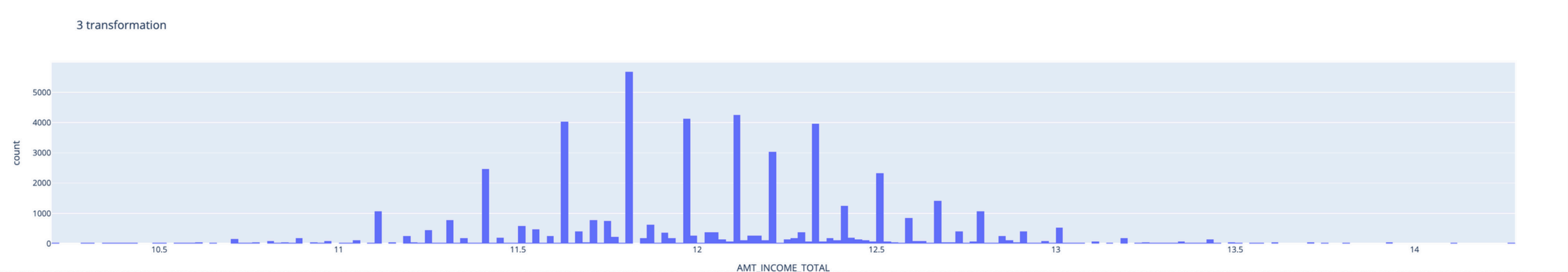
☒ Log Transformation

☐ Square Root Transformation

Submit

Data Preview (Random Sample 10 row)

| CNT_CHILDREN | AMT_INCOME_TOTAL | DAYS_BIRTH | DAYS_EMPLOYED | CNT_FAM_MEMBERS | AGE |
|--------------|------------------|------------|---------------|-----------------|-----|
| 0            | 13.42            | -17964     | -6594         | 2               | 49  |
| 1            | 12.51            | -15608     | -2755         | 3               | 43  |
| 0            | 11.81            | -20045     | -4335         | 2               | 55  |
| 0            | 12.32            | -20298     | -3282         | 2               | 56  |
| 0            | 12.1             | -22268     | 365243        | 2               | 61  |
| 0            | 12.32            | -20989     | 365243        | 2               | 58  |
| 0            | 12.42            | -16610     | -5221         | 2               | 46  |
| 0            | 13.02            | -19150     | -1511         | 2               | 52  |
| 1            | 11.97            | -12676     | -1350         | 2               | 35  |
| 0            | 12.51            | -12894     | -369          | 2               | 35  |



 Credit card Approval

Data Cleaning

Outlier detection and removal

Dimensionality Seduction(PTA)

Normality tests

Data Transformation

Loading Data

Plots

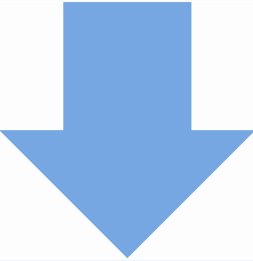
file upload

Drag and Drop or Select Files

Data Preview

Download CSV





 Credit card Approval

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Dimensionality Seduction(PTA)

Normality tests

Data Transformation

Loading Data

Plots

file upload

Drag and Drop or Select Files

Data Preview

Download CSV

| Unnamed: 0.1 | Unnamed: 0 | month_number | facecream | facewash | toothpaste | bathingsoap | shampoo | moisturizer | total_units | total_profit |
|--------------|------------|--------------|-----------|----------|------------|-------------|---------|-------------|-------------|--------------|
| 0            | 0          | 1            | 2500      | 1500     | 5200       | 9200        | 1200    | 1500        | 21100       | 211000       |
| 1            | 1          | 2            | 2630      | 1200     | 5100       | 6100        | 2100    | 1200        | 18330       | 183300       |
| 2            | 2          | 3            | 2140      | 1340     | 4550       | 9550        | 3550    | 1340        | 22470       | 224700       |
| 3            | 3          | 4            | 3400      | 1130     | 5870       | 8870        | 1870    | 1130        | 22270       | 222700       |
| 4            | 4          | 5            | 3600      | 1740     | 4560       | 7760        | 1560    | 1740        | 20960       | 209600       |
| 5            | 5          | 6            | 2760      | 1555     | 4890       | 7490        | 1890    | 1555        | 20140       | 201400       |
| 6            | 6          | 7            | 2980      | 1120     | 4780       | 8980        | 1780    | 1120        | 29550       | 295500       |
| 7            | 7          | 8            | 3700      | 1400     | 5860       | 9960        | 2860    | 1400        | 36140       | 361400       |
| 8            | 8          | 9            | 3540      | 1780     | 6100       | 8100        | 2100    | 1780        | 23400       | 234000       |
| 9            | 9          | 10           | 1990      | 1890     | 8300       | 10300       | 2300    | 1890        | 26670       | 266700       |
| 10           | 10         | 11           | 2340      | 2100     | 7300       | 13300       | 2400    | 2100        | 41280       | 412800       |
| 11           | 11         | 12           | 2900      | 1760     | 7400       | 14400       | 1800    | 1760        | 30020       | 300200       |

This page is for plotting graphs. Choose columns and plots to make.

Choose column x axis :

AGE

×

▼

Choose column y axis :

AMT\_INCOME\_TOTAL

×

▼

Choose Plot :

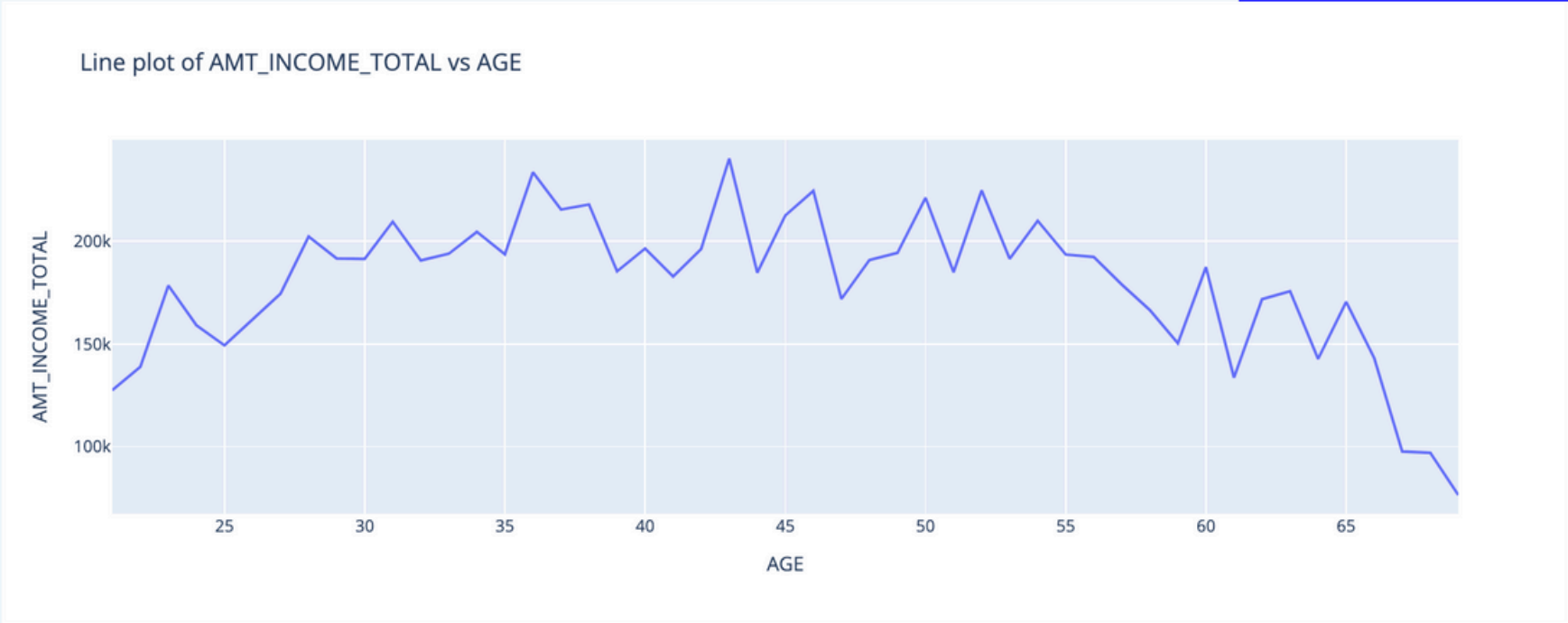
- ☒ Line plot
- ☐ Count plot
- ☐ Dist plot (Need 1 column)
- ☐ Heatmap with cbar
- ☐ Area plot
- ☐ Strip plot

Choose row size of the data :

0

10000

Submit



This page is for plotting graphs. Choose columns and plots to make.

Submit

URL : <https://dashapp-868428253622.northamerica-northeast1.run.app/>